Highlights

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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- Closed-loop knowledge-guided swarm learning for high-D feature selection.
- MI seeding with adaptive recovery prevents stagnation and restores diversity.
- Knowledge-guided elite refinement accumulates feature signals with decay.
- Two-phase weight scheduling decouples feature pruning from accuracy tuning.
- KDS-EL achieves 95.74% accuracy with only 87.8 features on 20 benchmarks.

Multi-Task Particle Swarm Optimization with Knowledge-Guided Elite Refinement and Two-Phase Weight Scheduling for High-Dimensional Feature Selection

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ABSTRACT

Feature selection in ultra-high-dimensional and sample-scarce regimes presents severe combinatorial complexity and heightened risk of classifier overfitting. Evolutionary swarm algorithms encounter three structural search degenerations: undirected initialization provokes cold-start evaluation waste in sparse feature spaces, memoryless uniform perturbation fails to harvest cross-iteration feature feedback, and static objective weighting causes a fundamental phase misalignment across evolutionary stages. To overcome these bottlenecks, this paper proposes Knowledge-Guided Dual-Swarm Evolutionary Learning (KDS-EL), a closed-loop framework for cooperative multi-task swarms. Specifically, mutual-information prior seeding and adaptive recovery eliminate cold-start inefficiency while preserving swarm diversity. A differential importance memory model with exponential forgetting guides targeted elite refinement within strictly bounded evaluations. Furthermore, dynamic two-phase weight scheduling temporally decouples early dimensional pruning from late classification refinement. Extensive evaluations across 20 high-dimensional benchmark datasets demonstrate that KDS-EL establishes comprehensive leadership under multi-criteria evaluation, achieving an overall mean classification accuracy of 95.74% while isolating an ultra-sparse signature averaging merely 87.8 features and delivering a 3.4-fold acceleration over the original MEL algorithm. Non-parametric Friedman rankings and post-hoc statistical tests corroborate its omnibus performance superiority across heterogeneous problem domains.

1. Introduction

High-dimensional data present fundamental challenges to pattern classification across diverse domains, including bioinformatics, transcriptomics, and biomedical diagnostics. These domains frequently involve datasets containing thousands or tens of thousands of continuous features alongside severely restricted sample sizes, creating ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly exceed available observational samples $N_{\text{samp}} \ll 100$. Under such adverse conditions, conventional machine learning algorithms suffer from severe overfitting, prohibitive computational costs, and degraded model interpretability due to the curse of dimensionality [1, 2]. Feature selection has consequently emerged as an indispensable preprocessing technique aimed at identifying a minimal, highly discriminative subset of relevant features while eliminating redundant and uninformative dimensions. Nevertheless, discovering the optimal biomarker subset is an intrinsically combinatorial problem whose discrete search space scales exponentially as 2^D, rendering exhaustive evaluation NP-hard [3]. Furthermore, univariate feature ranking criteria evaluate dimensions in isolation, fundamentally failing to capture multi-feature synergy and non-linear interactions essential for accurate classification.

Feature selection methods are broadly categorized into filter, wrapper, and embedded paradigms [1, 4]. Filter methods evaluate candidate features based on intrinsic statistical or information-theoretic metrics independently of inductive classifiers, offering high computational speed but frequently overlooking complex multi-feature synergy. Embedded methods incorporate feature selection directly into learning model parameter estimation, providing intermediate computational efficiency; however, their selection performance remains architecturally tied to specific classifiers. Conversely, wrapper methods directly guide feature subset exploration using classifier predictive feedback, naturally capturing non-linear feature interactions and yielding superior classification accuracy, albeit at the expense of intensive computational evaluation. To navigate the combinatorial discrete search space of wrapper feature selection, evolutionary computation has been extensively investigated as a powerful population-based optimization paradigm [3, 26]. By maintaining a diverse population of candidate solutions, evolutionary algorithms explore the discrete combinatorial space without requiring gradient information or restrictive convexity assumptions. Among various evolutionary meta-heuristics, Particle Swarm Optimization (PSO) has demonstrated prominent utility in feature selection tasks due to its conceptual simplicity, rapid information-sharing dynamics, and flexible velocity-position representation [5–7].

Despite its widespread adoption, canonical PSO encounters three critical structural bottlenecks when navigating ultra-high-dimensional and sample-scarce search spaces. First, standard uniform random initialization scatters particles indiscriminately across the candidate space, whereas optimal biomarker subsets occupy an ultra-sparse combinatorial subspace with cardinality $d^\star \ll D$. Consequently, a substantial volume of expensive wrapper evaluations is squandered on dense, uninformative candidate masks, severely exacerbating cold-start latency and sample-scarce overfitting. Second, conventional uniform variation heuristics treat all candidate dimensions equally and operate memorylessly across generations; by discarding the historical feature-level fitness feedback accumulated during previous iterations, the swarm fails to exploit verified discriminative dimensions and degenerates into stochastic random drift. Third, existing evolutionary wrappers enforce static, fixed objective weights to balance classification accuracy against subset cardinality throughout evolution. This static formulation creates a fundamental evolutionary phase misalignment: the search intrinsically demands aggressive dimensional pruning during early exploration to eliminate vast noise subspaces, whereas fine-grained decision-boundary tuning is essential during late exploitation to maximize inductive classification accuracy.

Recent evolutionary feature selection studies have investigated specialized strategies to mitigate high-dimensional search degradation, including feature-weight velocity steering in FWPSO [8], adaptive differential evolution weighting in SaWDE [9], surrogate-guided perturbation in IGPSO [10], and multi-criterion evaluation in PSO-CSM [11]. Although these single-objective approaches yield valuable localized improvements, they address initialization, perturbation, and weighting in isolation, lacking a unified closed-loop architecture. Concurrently, multi-task evolutionary optimization has emerged as a promising paradigm for scaling feature selection to massive dimensions. Most notably, the foundational Multi-Task Evolutionary Learning (MEL) framework [12] established a collaborative dual-subpopulation architecture to search complementary continuous and discrete representations. Nevertheless, existing multi-task frameworks leave critical structural vulnerabilities unresolved: continuous particles remain uniformly scattered without domain prior guidance, empirical feature-importance accumulation lacks exponential forgetting and elitist verification, and static objective scalarization persists throughout evolution.

To resolve these collective limitations, this paper proposes Knowledge-Guided Dual-Swarm Evolutionary Learning (KDS-EL), a closed-loop framework that cohesively integrates domain prior injection, cross-iteration knowledge accumulation, and dynamic objective scheduling into a cooperative multi-task PSO backbone.

The main contributions of this paper are summarized as follows:

1. *Prior-Guided Sparse Initialization with Adaptive Stagnation Recovery*: A hybrid seeding strategy that embeds continuous mutual-information soft priors into continuous decision variables without irreversible dimensional truncation, preserving full search-space accessibility while integrating Gaussian dispersion and an adaptive stagnation recovery mechanism to eliminate cold-start evaluation waste without sacrificing diversity (Section 3.2).

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Table 1: Methodological taxonomy of investigated evolutionary feature selection algorithms.

		Knowledge Engineering			Search Control		
Algorithm	Paradigm	MI Prior Seeding	Online Memory	Memory Decay	Elite Refinement	Dynamic Weighting	Complexity
BPSO [7]	SO-SI	×	×	×	×	×	High
FWPSO [8]	SO-SI	○	×	×	×	×	Moderate
SaWDE [9]	SO-DE	×	×	×	×	○	High
IGPSO [10]	SO-SI	○	×	×	×	×	Moderate
PSO-CSM [11]	SO-SI	○	×	×	×	×	Moderate
DENCA-SO [14]	SO-DE	○	×	×	×	×	Low
bIRIME [13]	SO-SI	×	×	×	○	×	High
BinCOA [15]	SO-SI	×	×	×	○	×	High
MEL [12]	MT-SI	×	✓	×	×	×	High
KDS-EL (Proposed)	**MT-SI**	✓	✓	✓	✓	✓	**Low**

✓: Fully supported; ○: Partially supported; ×: Absent / not supported.
Paradigms: SO: Single-Objective; MT: Multi-Task; SI: Swarm Intelligence; DE: Differential Evolution.

2. *Knowledge-Guided Elite Refinement (KER)*: A targeted elite refinement strategy that dynamically harvests cross-iteration feature-importance feedback with exponential memory decay, directing elite solutions toward historically effective dimensions within a strictly bounded evaluation overhead (Section 3.3).
3. *Two-Phase Weight Scheduling (TPS)*: A stage-aware objective scheduling mechanism that temporally decouples early aggressive feature pruning from late-stage accuracy refinement, overcoming the rigidity of static weighting (Section 3.4).
4. *Comprehensive Empirical Benchmark Validation*: Extensive evaluations across 20 high-dimensional benchmark datasets spanning 500 to 54,675 features against state-of-the-art single-objective competitor algorithms and the original MEL algorithm, rigorously validated through non-parametric statistical testing, component contribution analysis, and parametric sensitivity investigations (Section 4).

Extensive empirical evaluations across 20 high-dimensional benchmark datasets demonstrate that KDS-EL achieves decisive leadership among evolutionary wrapper feature selection paradigms under comprehensive multi-criteria evaluation, securing the highest overall mean classification accuracy alongside unprecedented feature parsimony. Specifically, KDS-EL attains an omnibus mean accuracy of 0.9574 while isolating an ultra-compact biomarker subset averaging merely 87.8 active features, representing an active dimension compression exceeding 98%. Furthermore, despite evaluating inductive wrapper classification boundaries, KDS-EL achieves a 3.4-fold empirical acceleration over the original MEL algorithm [12]. Non-parametric Friedman rankings and post-hoc Holm–Bonferroni step-down verification confirm that KDS-EL establishes statistically robust global leadership across the benchmark suite, achieving an optimal trade-off between predictive discrimination, extreme dimensional parsimony, and runtime efficiency.

The remainder of this paper is organized as follows. Section 2 formulates the feature selection optimization problem and reviews related literature and background concepts. Section 3 presents the proposed KDS-EL framework and formulates its three constituent mechanisms in detail. Section 4 presents comprehensive experimental results, statistical tests, component contribution analysis, and parameter sensitivity investigations. Finally, Section 5 concludes the paper and discusses prospective research directions.

2. Related Work and Background

This section establishes the formal mathematical foundation of feature selection and systematically reviews relevant evolutionary computation paradigms. Section 2.1 first formalizes the combinatorial optimization problem and search taxonomy. Section 2.2 analyzes evolutionary swarm dynamics alongside the structural failure modes of discrete binarization in high-dimensional spaces. Section 2.3 scrutinizes the cooperative multi-task evolutionary learning framework and pinpoints its operational bottlenecks. Finally, Section 2.4 reviews knowledge-guided evolutionary search and establishes the taxonomic positioning of KDS-EL across contemporary literature.

2.1. Problem formulation and search paradigms in feature selection

The essence of feature selection lies in combinatorial optimization, aiming to identify an optimal, minimal subset of informative features from a high-dimensional feature space to maximize classification performance while eliminating redundant and uninformative dimensions [1, 2]. Formally, let $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\text{samp}}}$ denote a supervised training dataset comprising N_{samp} instances, where $\mathbf{f}_i \in \mathbb{R}^D$ represents the full feature vector in the original D-dimensional space and $y_i \in \{1, 2, \ldots, C\}$ designates the discrete class label across C categories. In discrete binary representations, a candidate feature subset is formulated as a binary decision indicator vector $\mathbf{x} \in \{0, 1\}^D$, defined as:

$$x_j = \begin{cases} 1, & \text{if feature } f_j \text{ is selected,} \\ 0, & \text{if feature } f_j \text{ is discarded.} \end{cases} \tag{1}$$

The primary objective of feature selection is to simultaneously minimize the classification error rate and the cardinality of the selected feature subset. In evolutionary wrapper feature selection, this multi-criterion goal is mathematically formalized as minimizing a scalarized single-objective fitness function $f(\mathbf{x})$ [16, 17]:

$$\min_{\mathbf{x}\in\{0,1\}^D} f(\mathbf{x}) = w_a \cdot \text{ErrorRate}(\mathbf{x}) + w_f \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{2}$$

subject to $\mathbf{x} \in \{0, 1\}^D$ and $w_a + w_f = 1$, where $\text{ErrorRate}(\mathbf{x})$ represents the cross-validation classification error evaluated on the masked feature subset $\{\mathbf{f}_i \odot \mathbf{x}\}$ using an inductive k-nearest neighbors (k-NN) learner, with $\odot$ denoting the Hadamard element-wise product. The term $\|\mathbf{x}\|_0$ denotes the number of active features, $\|\mathbf{x}\|_0/D$ represents the feature selection ratio, and $w_a, w_f \in [0, 1]$ serve as trade-off coefficients balancing predictive accuracy against subset compactness. Because the cardinality of the discrete search space is 2^D, evaluating all candidate subsets exhaustively is NP-hard and computationally intractable when $D \gg 10^3$. Traditional feature selection techniques are broadly categorized into three classical paradigms based on their evaluation criteria: filter, embedded, and wrapper methods [2, 4]. Specifically, *filter methods* evaluate feature subsets based solely on intrinsic statistical or information-theoretic metrics—such as mutual information in mRMR [18], correlation in FCBF [19], or centrality scores [20]—independently of any classification algorithm. While filter methods are computationally lightweight with $O(DN_{\text{samp}} + D \log D)$ complexity, their classifier-agnostic evaluation and rigid hard truncation permanently discard weakly relevant features that exhibit strong collective synergy when combined. Conversely, *embedded methods* perform feature selection natively during classifier training by embedding sparsity constraints into model parameter estimation, such as L_1-regularized regression in Bolasso [21] or tree-based ensemble importance in FS-GBDT [22]. Although embedded methods bridge computational efficiency and predictive feedback, they remain architecturally bound to specific learners and frequently encounter performance degradation under severe sample scarcity where $N_{\text{samp}} \ll 100$. In contrast, *wrapper methods* utilize a target learning algorithm as a black-box evaluator, assessing candidate feature masks directly through cross-validation predictive error feedback [4, 16]. By tightly coupling subset evaluation with classifier feedback, wrappers naturally capture complex non-linear feature interactions and yield superior classification accuracy.

To navigate the discrete 2^D combinatorial space in wrapper feature selection without gradient information, Evolutionary Computation paradigms—including Genetic Algorithms [23], Ant Colony Optimization [24], Artificial Bee Colony [25], and Swarm Intelligence [3, 26]—have emerged as dominant search heuristics. To combat the curse of dimensionality in small-sample high-dimensional data, various advanced single-objective evolutionary strategies have been introduced. For instance, self-adaptive differential evolution with adaptive weighting (SaWDE) [9] dynamically balances classification error against subset cardinality; feature-weight particle swarm optimization (FWPSO) [8] injects offline feature weights to steer velocity updates; importance-guided particle swarm optimization (IGPSO) [10] employs multi-layer perceptron importance scores to bias particle movements; and particle swarm optimization with comprehensive scoring mechanism (PSO-CSM) [11] leverages multi-criterion feature evaluation to restrict search trajectories. Despite these advancements, standard single-objective evolutionary wrappers face severe structural bottlenecks in ultra-high-dimensional spaces where $D \gg 10^3$. Uniform random initialization scatters individuals indiscriminately across the search space, generating overly dense candidate masks with approximately $D/2$ active features that burden classifier evaluations and provoke premature overfitting. Furthermore, stochastic variation operators treat feature perturbations as memoryless trials, discarding historical feature-importance feedback across generations. Finally, static or coarsely tuned objective weighting enforces rigid trade-offs throughout evolution, failing to adapt to the distinct requirements of early-stage dimensional reduction and late-stage accuracy refinement.

2.2. Evolutionary swarm intelligence and discrete feature selection dynamics

Standard particle swarm optimization, originally introduced by Kennedy and Eberhart [5], is a population-based stochastic meta-heuristic inspired by the collective foraging behaviors of bird flocks and fish schools. A swarm consists of N candidate particles navigating a D-dimensional continuous solution space $\mathbb{R}^D$. At iteration t, each particle i is characterized by a continuous position vector $\mathbf{p}_i^{(t)} \in \mathbb{R}^D$ and an associated velocity vector $\mathbf{v}_i^{(t)} \in \mathbb{R}^D$, steered dynamically by its personal historical best position $\mathbf{p}_{\text{best},i} \in \mathbb{R}^D$ and the swarm's global best position $\mathbf{g}_{\text{best}} \in \mathbb{R}^D$. The velocity and position updates of particle i at iteration $t+1$ are formulated as follows [6]:

$$\mathbf{v}_i^{(t+1)} = \omega \mathbf{v}_i^{(t)} + c_1 r_1 (\mathbf{p}_{\text{best},i} - \mathbf{p}_i^{(t)}) + c_2 r_2 (\mathbf{g}_{\text{best}} - \mathbf{p}_i^{(t)}), \tag{3}$$

$$\mathbf{p}_i^{(t+1)} = \mathbf{p}_i^{(t)} + \mathbf{v}_i^{(t+1)}, \tag{4}$$

where $\omega \in [0, 1]$ denotes the inertia weight controlling momentum retention, $c_1, c_2 > 0$ are the cognitive and social acceleration coefficients governing the attraction toward individual and collective best experiences, and $r_1, r_2 \sim \mathcal{U}(0, 1)$ are independent uniform random factors providing stochastic exploration capability.

To adapt continuous PSO to the combinatorial nature of binary feature selection where candidate solutions are represented as binary indicator masks $\mathbf{x}_i \in \{0, 1\}^D$, Kennedy and Eberhart [7] proposed Binary PSO (BPSO). In BPSO, continuous velocities are mapped into selection probabilities via a Sigmoid transfer function:

$$S(v_{i,j}^{(t+1)}) = \frac{1}{1 + \exp(-v_{i,j}^{(t+1)})}, \tag{5}$$

and the corresponding binary decision variable $x_{i,j}^{(t+1)}$ is sampled probabilistically as:

$$x_{i,j}^{(t+1)} = \begin{cases} 1, & \text{if } r_{i,j} < S(v_{i,j}^{(t+1)}), \\ 0, & \text{otherwise}, \end{cases} \tag{6}$$

where $r_{i,j} \sim \mathcal{U}(0, 1)$ is an independent uniform random threshold. Although BPSO exhibits fast convergence on low-dimensional benchmarks, it encounters severe exploration-exploitation dilemmas when applied to ultra-high-dimensional biomedical domains where $D \gg 10^3$. Because the Sigmoid function satisfies $S(0) = 0.5$, continuous velocities hovering near zero inevitably yield an expected active feature density of approximately $D/2$. In typical biomedical datasets containing $D \geq 10{,}000$ dimensions, this stochastic mapping generates candidate subsets with more than 5,000 active features, heavily overloading wrapper classifier evaluations with $O(N_{\text{samp}}^2 d_{\text{selected}})$ complexity and triggering severe overfitting under small sample sizes where $N_{\text{samp}} \ll 100$. Furthermore, continuous stochastic sampling introduces high generational variance: even when velocity vectors stabilize, candidate feature masks continue to jitter stochastically, causing particle trajectories to degrade into memoryless random walks across the combinatorial space.

To overcome the random drift of standard binarization in high-dimensional feature selection, diverse single-objective meta-heuristic algorithms have recently been proposed. Evolutionary filter methods, such as DENCA-SO [14], employ neighborhood component feature selection (NCFS) continuous weights within differential evolution operators. By evaluating candidate subsets using algebraic surrogate costs rather than wrapper classifiers, DENCA-SO achieves rapid wall-clock execution; however, because its optimization process remains entirely decoupled from inductive classification margins, it suffers from severe feature redundancy and retains over 48% of original dimensions on biomedical benchmarks. Concurrently, advanced binary swarm optimizers seek to enhance discrete exploration. For instance, bIRIME [13] incorporates dynamic escape energy, composite mutation strategies, and opposition-based restart mechanisms to mitigate premature convergence, whereas BinCOA [15] couples refractive opposition-based learning with crossed horizontal and vertical search operations. Nevertheless, these multi-operator strategies incur prohibitive computational burdens: generating and evaluating multiple trial vectors per agent scales wrapper classifier calls to three or four times the population size per generation, causing execution times to escalate beyond 2,000 seconds on large-sample benchmarks (Section 4.2.6). Crucially, in ultra-high dimensions without proactive pruning guidance, stochastic binarization traps these algorithms near the uninformative 50% dimensionality attractor, preserving thousands of redundant features. On the other hand, wrapper methods employing scalarized fitness formulations—such as SaWDE [9], FWPSO [8], and PSO-CSM [11]—enforce fixed static objective weights throughout evolution. This static weighting creates a fundamental evolutionary phase misalignment: the search process inherently requires aggressive feature pruning during early exploration to eliminate vast noise subspaces, whereas delicate decision-boundary fine-tuning is required during late-stage exploitation to maximize predictive accuracy.

2.3. Multi-task evolutionary learning and the MEL framework

Evolutionary Multi-Task Optimization (EMTO) solves multiple distinct optimization tasks simultaneously by facilitating inter-task knowledge transfer across unified representations [27]. In feature selection, EMTO frameworks divide the population into collaborative subpopulations exploring complementary search subspaces [28–32]. Most notably, Wang et al. [12] introduced Multi-Task Evolutionary Learning (MEL), establishing a dual-subpopulation cooperative framework. In the original MEL algorithm, a population of N particles is partitioned into collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, each comprising $N/2$ individuals operating on complementary representations. Specifically, *Subpopulation 1* executes continuous swarm exploration in $[0, 1]^D$, updating particle velocities and positions through PSO dynamics augmented with an inter-task knowledge transfer channel directing particles toward the historical global best position $\mathbf{g}_{\text{best},2} \in \{0, 1\}^D$ of Subpopulation 2:

$$\begin{aligned} \mathbf{v}_{1,i}^{(t+1)} = \omega \mathbf{v}_{1,i}^{(t)} &+ c_1 \mathbf{r}_1 \odot (\mathbf{p}_{\text{best},1,i} - \mathbf{x}_{1,i}^{(t)}) \\ &+ c_2 \mathbf{r}_2 \odot (\mathbf{g}_{\text{best},1} - \mathbf{x}_{1,i}^{(t)}) + c_3 \mathbf{r}_3 \odot (\mathbf{g}_{\text{best},2} - \mathbf{x}_{1,i}^{(t)}), \end{aligned} \tag{7}$$

where $\mathbf{x}_{1,i}^{(t)} \in [0, 1]^D$ and $\mathbf{v}_{1,i}^{(t)} \in [v_{\min}, v_{\max}]^D$ denote the continuous position and velocity of particle i, $\mathbf{p}_{\text{best},1,i}$ and $\mathbf{g}_{\text{best},1}$ represent personal and local bests, $\omega \in [0, 1]$ is the inertia weight, $c_1, c_2, c_3 > 0$ are acceleration coefficients, $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3 \sim \mathcal{U}(0, 1)^D$, and $\odot$ denotes the Hadamard product. Continuous positions are binarized into candidate selection masks via transfer mapping. Concurrently, *Subpopulation 2* performs discrete sampling from a shared importance matrix $\mathbf{W}_{\text{MEL}} \in \mathbb{R}^D$ accumulating historical feature selection frequencies across candidate solutions. Both subpopulations evaluate candidate subsets via a shared scalar fitness function similar to Eq. (2) with fixed static weights $\alpha, \beta \in [0, 1]$ satisfying $\alpha + \beta = 1$.

While MEL establishes the utility of cross-task knowledge transfer, its search dynamics encounter critical structural bottlenecks in ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly exceed observations $N_{\text{samp}} \ll 100$. First, continuous particles in $\mathcal{P}_1$ are initialized uniformly without domain guidance, scattering early exploration and squandering wrapper evaluations across uninformative subspaces, motivating the prior initialization and adaptive recovery strategies in Section 3.2. Second, matrix $\mathbf{W}_{\text{MEL}}$ accumulates historical selection frequencies without exponential forgetting or elite fitness verification, prematurely biasing discrete sampling in $\mathcal{P}_2$ and necessitating the differential importance learning and targeted elite refinement mechanisms in Section 3.3. Third, enforcing static objective weights creates an evolutionary phase misalignment between early coarse pruning and late accuracy refinement, resolved by the dynamic Two-Phase Weight Scheduling strategy in Section 3.4. In response, KDS-EL establishes a closed-loop evolutionary learning framework tailored for high-dimensional feature selection.

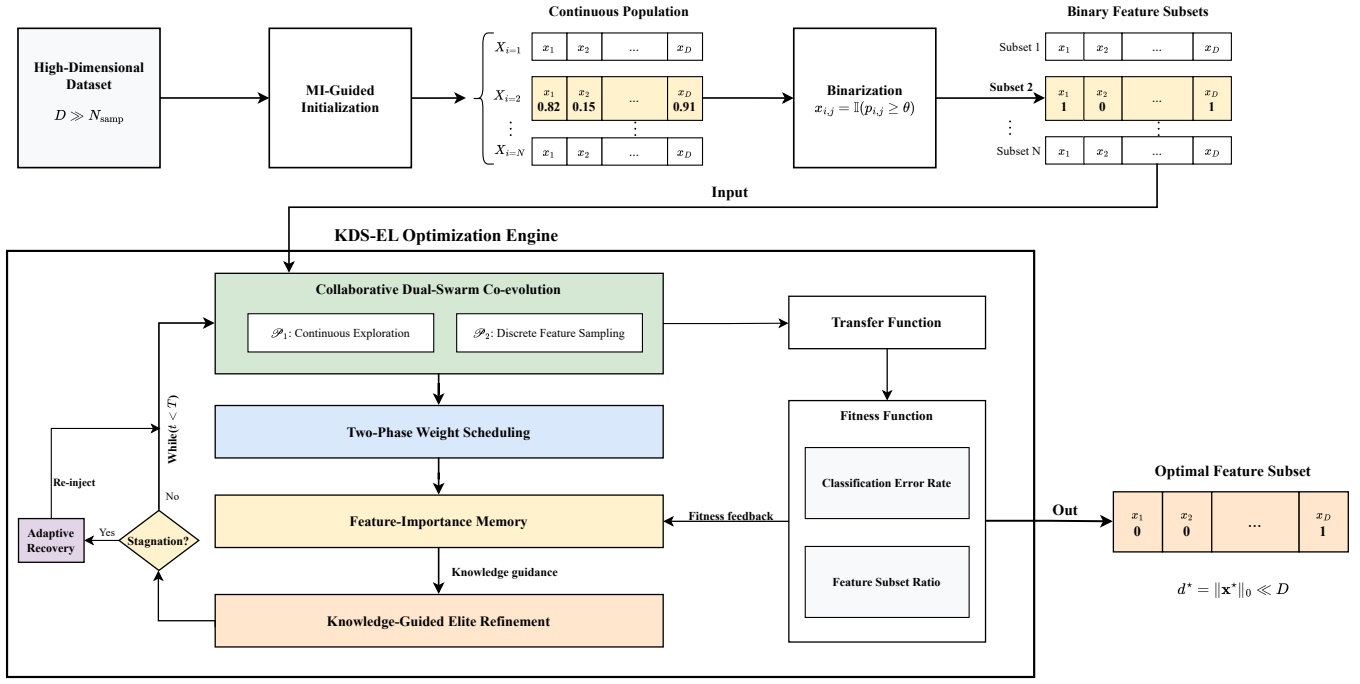


Figure 1: Overall dataflow architecture of the proposed KDS-EL framework for high-dimensional feature selection.

2.4. Knowledge-guided search and taxonomic positioning

Integrating domain knowledge into meta-heuristics effectively mitigates the blindness of black-box combinatorial search [17, 33, 34]. In high-dimensional feature selection, Mutual Information (MI), rooted in Shannon's information theory [35], quantifies non-linear dependency between continuous feature variables $f_j \in \mathbb{R}$ and discrete class labels $Y \in \{1, \ldots, C\}$:

$$\begin{aligned} I(f_j; Y) &= H(f_j) - H(f_j \mid Y) \\ &= \sum_{c=1}^{C} P(Y = c) \int_{\mathbb{R}} p(f_j \mid Y = c) \log_2 \frac{p(f_j \mid Y = c)}{p(f_j)} \, df_j, \end{aligned} \tag{8}$$

where $P(Y = c)$ is the class prior probability, while $p(f_j)$ and $p(f_j \mid Y = c)$ denote the marginal and conditional density functions of continuous feature f_j. To avoid bin-width sensitivity in empirical applications with unknown distributions, non-parametric nearest-neighbor estimators—specifically the Kraskov–Stögbauer–Grassberger (KSG) estimator [36] extended by Ross [37]—directly evaluate non-linear dependencies. Crucially, in ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly outnumber observations $N_{\text{samp}} \ll 100$, continuous estimators inevitably incur finite-sample estimation variance, producing ranking noise. Conventional filter methods [17, 18] evaluate mutual information in $O(DN_{\text{samp}} + D \log D)$ time but enforce rigid hard truncation, retaining only the top-k ranked features. This hard cut-off fundamentally ignores multi-feature synergy: weakly individual biomarkers that are collectively decisive when combined are permanently eliminated, while false positives elevated by finite-sample noise are irreversibly locked into the candidate pool.

Existing knowledge-guided evolutionary methods primarily adopt one-shot offline injection via either static subspace pre-selection [17, 34] or static velocity biasing [8]. While beneficial in narrowing early search, one-shot injection lacks online adaptability: the static prior remains frozen and cannot assimilate iterative trial-and-error feedback from inductive classification margins. Conversely, online surrogate-assisted approaches—such as multi-layer perceptrons [10] or Gaussian processes [38]—attempt dynamic importance learning but incur prohibitive retraining latencies and suffer severe model misspecification under sample-scarce regimes where observations $N_{\text{samp}} \ll 100$ cannot constrain high-capacity surrogates.

To resolve the limitations of rigid static priors and heavyweight surrogates, KDS-EL establishes a closed-loop, knowledge-guided evolutionary learning framework unifying offline domain priors, online memory harvesting, and stage-aware objective scheduling. Specifically, normalized mutual information scores are embedded as continuous Gaussian-perturbed soft seeds, preserving the complete search space while steering the swarm toward an ultra-sparse initial basin, complemented by an adaptive recovery mechanism to restore diversity upon stagnation (Section 3.2). Throughout evolution, cross-iteration fitness feedback is accumulated into a differential importance matrix $\mathbf{W} \in \mathbb{R}^D$ governed by exponential forgetting, enabling the KER strategy to execute targeted, directional elite flips within a strictly bounded evaluation budget (Section 3.3). Finally, a dynamic Two-Phase Weight Scheduling strategy temporally decouples early coarse pruning from late boundary refinement, achieving simultaneous gains in classification accuracy, feature compactness, and runtime efficiency (Section 3.4). Table 1 systematically summarizes the methodological taxonomy and structural comparison between KDS-EL and all nine competitor feature selection algorithms investigated in this study across seven orthogonal technical dimensions.

3. The Proposed Method

This section details the theoretical foundations and algorithmic architecture of the proposed KDS-EL framework. Section 3.1 first outlines the overarching data-flow architecture and continuous-discrete dual decision space. Section 3.2 formulates the prior-guided sparse population initialization and adaptive stagnation recovery mechanisms. Section 3.3 establishes the closed-loop differential importance memory model and knowledge-guided elite refinement strategy. Section 3.4 develops the dynamic Two-Phase Weight Scheduling strategy. Finally, Section 3.5 synthesizes the overall algorithmic coordination alongside asymptotic computational complexity analysis.

3.1. The framework of KDS-EL

The macro data-flow architecture and problem mapping of KDS-EL for high-dimensional feature selection are systematically illustrated in Fig. 1. Given a high-dimensional dataset $\mathcal{D}$ characterized by severe sample scarcity, the search pipeline begins by extracting filter-level mutual information to construct a continuous prior relevance landscape $\mathbf{m} \in [0, 1]^D$. An initial population of continuous positions $\mathbf{p}_i \in [0, 1]^D$ is generated via hybrid prior seeding and deterministically mapped into sparse binary candidate masks $\mathbf{x}_i \in \{0, 1\}^D$ via Heaviside thresholding. Throughout optimization, these candidate subsets are transferred to the KDS-EL optimization engine, which coordinates collaborative dual-swarm co-evolution between subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, dynamic Two-Phase Weight Scheduling, wrapper classification evaluations, cumulative feature-importance harvesting into memory vector $\mathbf{W} \in \mathbb{R}^D$, and targeted elite refinement in closed-loop synergy. Upon satisfying the termination criterion, the framework isolates the optimal compact biomarker subset $\mathbf{x}^\star$ with cardinality $d^\star = \|\mathbf{x}^\star\|_0 \ll D$ alongside its cross-validated predictive performance.

The execution logic and operational flowchart across evolutionary generations are depicted in Fig. 2. In KDS-EL, candidate solutions operate under

a continuous-discrete dual representation: continuous coordinates reside in the unit hypercube $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^T \in [0,1]^{N\times D}$, while the evaluated feature subsets across D dimensions are formally represented as an $N \times D$ binary decision matrix $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_N \end{bmatrix} = \begin{bmatrix} x_{1,1} & x_{1,2} & \cdots & x_{1,D} \\ x_{2,1} & x_{2,2} & \cdots & x_{2,D} \\ \vdots & \vdots & \ddots & \vdots \\ x_{N,1} & x_{N,2} & \cdots & x_{N,D} \end{bmatrix} \in \{0,1\}^{N\times D}, \tag{9}$$

where the i-th row vector $\mathbf{x}_i = [x_{i,1}, x_{i,2}, \ldots, x_{i,D}] \in \{0,1\}^D$ denotes an individual candidate solution, with $x_{i,j} = 1$ indicating that feature f_j is selected and $x_{i,j} = 0$ indicating exclusion. The mapping from continuous particle position $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ to binary decision mask $\mathbf{x}_i$ is governed by a deterministic threshold condition $x_{i,j} = \mathbb{I}(p_{i,j} \geq \theta)$, where $\mathbb{I}(\cdot)$ denotes the indicator function and $\theta \in (0.5, 1)$ represents the binarization cut-off threshold.

To balance global exploration and local exploitation in sparse high-dimensional spaces, the swarm is partitioned into two collaborative subpopulations $\mathcal{P}_1$ and $\mathcal{P}_2$, each comprising $N/2$ particles, extending the multi-task evolutionary learning paradigm of the original MEL algorithm [12]. As delineated in the upper dashed sub-panel of Fig. 2, Subpopulation $\mathcal{P}_1$ functions as a continuous explorer, executing continuous velocity–position updates with an inter-task knowledge transfer channel guided by the global best solution $\mathbf{g}_{\text{best},2}$ of $\mathcal{P}_2$. Concurrently, Subpopulation $\mathcal{P}_2$ acts as a discrete exploiter, conducting discrete probability-driven feature sampling informed by the accumulated feature-importance landscape $\mathbf{W}$. Compared with conventional single-population architectures, this collaborative co-evolutionary design establishes complementary search dynamics: continuous swarm movement preserves momentum to traverse unvisited spaces, while discrete sampling focuses evaluations on verified high-relevance dimensions, effectively mitigating premature stagnation.

During iterative optimization, KDS-EL orchestrates three complementary mechanisms into an integrated closed loop. First, population initialization injects Gaussian-perturbed mutual information priors to establish an ultra-sparse candidate basin, while an adaptive re-initialization mechanism restores swarm diversity whenever search progress halts. Second, differential fitness improvements across generations are accumulated into a continuous importance vector $\mathbf{W} \in \mathbb{R}^D$ under exponential memory decay, actively steering the KER strategy to execute targeted directional bit flips under a greedy pairwise in-place replacement rule, as illustrated in the lower dashed sub-panel of Fig. 2. Third, the dynamic TPS strategy temporally shifts objective weights across evolutionary stages, enforcing aggressive feature pruning during the early phase and fine-grained accuracy tuning during the late phase. Architecturally, KDS-EL establishes an orthogonal spatio-temporal hierarchy: the collaborative dual-swarm structure governs spatial exploration across complementary continuous and discrete domains, whereas the dynamic TPS strategy orchestrates temporal search progression across decoupled pruning and refinement phases. By continuously cycling through these coordinated stages, KDS-EL systematically prunes non-informative dimensions while progressively refining compact biomarker signatures.

3.2. Population initialization and adaptive recovery

In ultra-high-dimensional wrapper feature selection, standard uniform random initialization samples continuous particle coordinates uniformly across the unit hypercube, activating each candidate feature with an expected probability of 0.50 and collapsing candidate subsets into the dense dimensionality attractor characterized by cardinality $d \approx D/2$. In high-dimensional regimes where $D \gg 10^3$, evaluating these dense candidate masks forces wrapper classifiers to compute pairwise distances across thousands of non-informative dimensions at a computational complexity scaling as $O(N_{\text{samp}}^2 d)$, which squanders early function evaluation budgets and precipitates severe classifier overfitting under limited sample observations where $N_{\text{samp}} \ll 100$. To establish an informative yet diverse initial swarm while systematically mitigating mid-evolution search stagnation, KDS-EL introduces a prior-guided sparse population initialization strategy coupled with an adaptive stagnation recovery mechanism, as systematically illustrated across the four operational stages in Fig. 3.

3.2.1. Prior-guided sparse population initialization

Given the training dataset instance $\mathcal{D} = \{(\mathbf{f}_k, y_k)\}_{k=1}^{N_{\text{samp}}}$ with continuous feature vectors $\mathbf{f}_k = [f_{k,1}, \ldots, f_{k,D}] \in \mathbb{R}^D$ and discrete class labels $y_k \in \{1, \ldots, C\}$, the non-linear dependencies between individual candidate dimensions and target categories are quantified through continuous-discrete mutual information [37]. To handle continuous real-valued inputs without relying on arbitrary data discretization, the mutual information score $\hat{I}(f_j; Y)$ for each feature $j \in \{1, \ldots, D\}$ is computed via the non-parametric k-nearest neighbors KSG estimator [36, 37]:

$$\hat{I}(f_j; Y) = \psi(N_{\text{samp}}) - \frac{1}{N_{\text{samp}}} \sum_{k=1}^{N_{\text{samp}}} \psi(N_{y_k}) + \psi(k_{\text{nn}}) - \frac{1}{N_{\text{samp}}} \sum_{k=1}^{N_{\text{samp}}} \psi(n_{f_j,k}), \tag{10}$$

where $\psi(\cdot)$ denotes the digamma function, N_{y_k} is the count of training samples sharing class label y_k, $k_{\text{nn}} = 3$ is the neighborhood parameter in the joint continuous-discrete space, and $n_{f_j,k}$ is the count of data points whose marginal continuous distance to sample k along feature f_j is strictly smaller than the distance to its k_{nn}-th neighbor in the class-conditioned subspace. To establish a standardized probability profile across heterogeneous datasets, these raw mutual information scores are normalized into a prior relevance vector $\mathbf{m} = [m_1, \ldots, m_D]^T \in [0,1]^D$:

$$m_j = \frac{\hat{I}(f_j; Y) - \min_{1\leq l\leq D} \hat{I}(f_l; Y)}{\max_{1\leq l\leq D} \hat{I}(f_l; Y) - \min_{1\leq l\leq D} \hat{I}(f_l; Y) + \epsilon}, \quad j = 1, \ldots, D, \tag{11}$$

where $\epsilon > 0$ is a small numerical constant preventing division by zero.

To guarantee strict methodological integrity and eliminate selection bias, the filter prior calculation is architecturally restricted to the training partition presented to the optimizer. In an inductive machine learning deployment where an external unseen test partition $\mathcal{D}_{\text{test}}$ or an outer nested evaluation loop is prescribed, the prior relevance vector $\mathbf{m}$ and the subsequent evolutionary wrapper search are executed strictly on the training partition $\mathcal{D}_{\text{train}}$, ensuring that unseen validation samples remain completely unobserved throughout all search stages. In the standardized benchmark optimization protocol adopted in this work (Section 4.1), $\mathcal{D}$ serves as the complete benchmark problem instance, within which an internal 5-fold cross-validation scheme functions as the wrapper fitness evaluation surrogate.

To prevent population homogenization while guiding search trajectories toward informative dimensions, candidate particles are initialized through a hybrid seeding scheme parameterized by a seeding ratio $\rho \in (0,1)$, as illustrated in Step 1 of Fig. 3. Continuous position vectors $\mathbf{p}_i = [p_{i,1}, \ldots, p_{i,D}] \in [0,1]^D$ are constructed as:

$$\mathbf{p}_i = \begin{cases} \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), & 1 \leq i \leq \lfloor \rho N \rfloor, \\ \mathbf{u}_i, & \lfloor \rho N \rfloor < i \leq N, \end{cases} \tag{12}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation σ, $\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ provides uniform stochastic coverage across unguided dimensions, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Continuous position vectors $\mathbf{p}_i$ are subsequently transformed into binary decision vectors $\mathbf{x}_i = [x_{i,1}, \ldots, x_{i,D}] \in \{0,1\}^D$ via a deterministic Heaviside step mapping with threshold $\theta \in (0.5, 1)$:

$$x_{i,j} = \begin{cases} 1, & \text{if } p_{i,j} \geq \theta, \\ 0, & \text{otherwise}, \end{cases} \quad j = 1, \ldots, D. \tag{13}$$

By setting the cut-off threshold $\theta > 0.50$, only features possessing above-average prior relevance are initialized in active states. This produces an intrinsically sparse initial population that biases exploration toward high-potential biomarker candidates, thereby resolving the cold-start bottleneck, reducing early evaluation latency, and accelerating convergence toward promising feature subspaces.

Estimation robustness and mitigation of the Winner's curse under sample scarcity: In ultra-high-dimensional and sample-scarce regimes where candidate dimensions $D \gg 10^3$ vastly exceed sample observations $N_{\text{samp}} \ll 100$ by two to three orders of magnitude, non-parametric entropy estimation on limited observations inherently exhibits non-trivial finite-sample variance. Under extensive dimensional testing, stochastic sample alignment across thousands of candidate features may cause irrelevant noise dimensions to obtain spuriously elevated mutual information estimates—a classic manifestation of the Winner's curse in high-dimensional screening. If subjected to standard Min-Max normalization, such estimation fluctuations risk being amplified toward $m_j \approx 1$, potentially injecting false-positive biases into the initial population.

Crucially, KDS-EL establishes a multi-layered architectural defense that insulates the evolutionary search against noisy prior contamination. First, KDS-EL fundamentally departs from conventional filter pre-selectors by rejecting irreversible hard feature truncation; rather than permanently discarding unselected dimensions, the normalized vector $\mathbf{m}$ establishes a continuous soft prior manifold, maintaining the complete $\mathbb{R}^D$ feature space fully accessible throughout optimization. Second, the hybrid initialization in (12) injects Gaussian dispersion $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ around $\mathbf{m}$ while maintaining an unguided sub-swarm

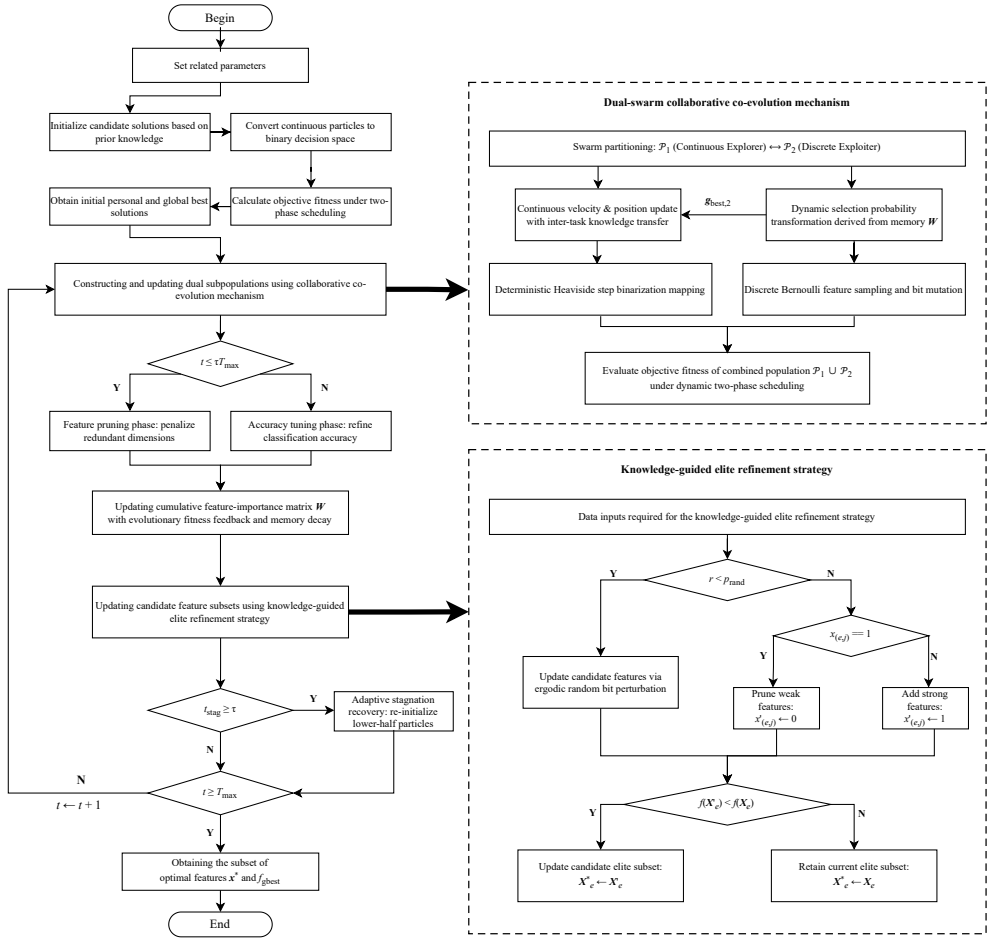


Figure 2: Algorithmic execution flowchart of the proposed KDS-EL framework.

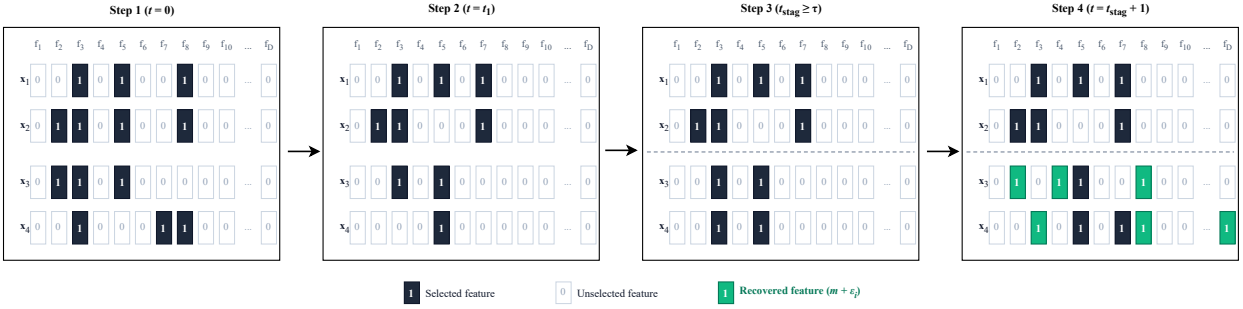


Figure 3: Schematic of prior-guided sparse population initialization and adaptive diversity recovery.

$\mathbf{u}_i \sim \mathcal{U}(0,1)^D$ across $(1-\rho)N$ particles, preventing premature population collapse into false-positive empirical peaks. Third, every candidate subset is immediately subjected to wrapper cross-validation classification feedback; because spurious noise features fail to generalize across cross-validation folds, they generate zero or negative fitness differentials. Under the exponential memory decay governed by decay factor γ in (17), their transient weights in the cumulative feature-importance vector $\mathbf{W} \in \mathbb{R}^D$ rapidly decay to zero. Consequently, the knowledge-guided elite refinement mechanism swiftly purges these non-contributing dimensions in favor of synergistic, predictive biomarkers. Finally, should the swarm become temporarily trapped in empirical local attractors, the adaptive stagnation recovery routine (Section 3.2.2) re-samples the stagnant subpopulation under fresh stochastic perturbations, ensuring robust long-term exploratory self-correction.

3.2.2. Adaptive stagnation recovery

In iterative swarm search over ultra-high-dimensional fitness landscapes, population diversity inevitably decays as particles converge toward shared historical attractors. As evolutionary generations progress, particle velocities decelerate toward zero, causing the swarm to cluster tightly within confined local basins. In high-dimensional regimes where candidate dimensions $D \gg 10^3$ vastly exceed sample observations $N_{\text{samp}} \ll 100$, such velocity decay precipitates premature search stagnation, trapping the optimizer in sub-optimal feature combinations governed by empirical noise. Conventional meta-heuristics frequently attempt to combat stagnation via uniform random re-initialization or heavy-tailed mutation perturbations, such as Cauchy mutations or opposition-based learning [13, 15]. However, in ultra-high-dimensional spaces, uniform random restarts inevitably collapse candidate solutions back into the dense dimensionality attractor characterized by cardinality $d \approx D/2$, squandering subsequent wrapper evaluations on non-informative noise dimensions. Similarly, blind heavy-tailed mutations in 2^D combinatorial space perturb candidate bit positions without domain guidance, dismantling carefully discovered, highly cooperative biomarker combinations with an overwhelming probability exceeding 99.9%. Furthermore, classical opposition-based learning strategies invert binary masks across all dimensions, transforming sparse subsets into massive complementary masks of cardinality $D - d \approx 10^4$, thereby exacerbating classifier overfitting and computational latency. Consequently, sustaining exploratory

diversity in ultra-high dimensions requires an asymmetric recovery mechanism capable of injecting targeted stochastic exploration while preserving verified elite solutions.

To resolve this structural dilemma without incurring catastrophic forgetting or dimensional bloat, KDS-EL introduces an adaptive stagnation recovery mechanism. The adaptiveness of this mechanism stems from its feedback-driven, on-demand activation: unlike classical heuristic restarts that disrupt search trajectories on fixed generational schedules regardless of convergence status, KDS-EL couples recovery execution dynamically to runtime fitness stagnation feedback. Throughout generational search, a stagnation counter t_{stag} monitors the consecutive generational span during which the global-best fitness fails to achieve a strict numerical decrease. As long as the swarm sustains productive descent, the counter t_{stag} is continuously reset, leaving ongoing trajectories undisturbed; recovery is triggered strictly on demand when $t_{\text{stag}} \geq \tau$, where $\tau \in \mathbb{N}^+$ serves as a preset patience parameter regulating detection sensitivity. When this condition is met, search stagnation is formally identified, triggering an asymmetric bipartite population bisection, as illustrated in Step 3 of Fig. 3. Specifically, candidate particles are ranked in ascending order according to their historical personal-best fitness values $f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t))$. To safeguard converged, highly discriminative biomarker combinations, the superior half of the swarm comprising $\lfloor N/2 \rfloor$ particles is preserved intact, maintaining the optimization memory accumulated along successful search trajectories. Concurrently, the lower-performing half comprising the remaining $N - \lfloor N/2 \rfloor$ stagnant particles has both its current continuous positions $\mathbf{p}_i$ and personal-best memory vectors $\mathbf{p}_{\text{best},i}$ refreshed by re-sampling around the continuous prior relevance vector $\mathbf{m} \in [0,1]^D$ under Gaussian perturbation:

$$\begin{aligned} &\mathbf{p}_i \leftarrow \text{clip}\left(\mathbf{m} + \boldsymbol{\varepsilon}_i,\ 0,\ 1\right), \quad \mathbf{p}_{\text{best},i} \leftarrow \mathbf{p}_i, \\ &x_{i,j} \leftarrow \mathbb{I}(p_{i,j} \geq \theta), \quad \forall j = 1, \ldots, D, \end{aligned} \tag{14}$$

where $\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_D)$ represents zero-mean Gaussian perturbation with standard deviation $\sigma \in \mathbb{R}^+$, $\mathbb{I}(\cdot)$ is the indicator function, $\theta \in (0.5, 1)$ denotes the cut-off binarization threshold, and $\text{clip}(z, 0, 1) = \max(0, \min(1, z))$ restricts continuous coordinates to the unit hypercube.

Immediately following position refreshing, the binarized candidate masks $\mathbf{x}_i$ of the refreshed subpopulation are evaluated via the internal cross-validation wrapper under the current phase objective weight vector $\mathbf{w}(t)$, as depicted in Step 4 of Fig. 3. The refreshed personal-best fitness records are updated to $f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) \leftarrow f(\mathbf{x}_i, \mathbf{w}(t))$, allowing newly sampled candidates to enter subsequent global-best competition alongside the preserved elite half, while the stagnation counter is reset to zero ($t_{\text{stag}} \leftarrow 0$). Crucially, this bipartite design strictly bounds the additional computational overhead to exactly $\lfloor N/2 \rfloor$ wrapper evaluations, which are invoked strictly on demand rather than continuously incurred across every generation. Furthermore, because re-sampling is anchored to the continuous prior relevance vector $\mathbf{m}$ rather than uniform noise, newly generated candidate masks remain intrinsically sparse, entirely circumventing the destructive $D/2$ dimensionality attractor. When coupled with the exponential memory forgetting of non-contributing dimensions in the cumulative feature-importance vector $\mathbf{W} \in \mathbb{R}^D$, this adaptive recovery mechanism breaks exploratory deadlocks, injects fresh variance into dormant subspaces, and propels the swarm toward globally competitive, sparse biomarker subsets.

3.3. Cumulative importance learning and elite refinement

To overcome the search amnesia and destructive perturbation bottlenecks of conventional meta-heuristics, KDS-EL establishes a closed-loop knowledge learning mechanism comprising online importance accumulation and targeted directional elite refinement.

In ultra-high-dimensional feature selection where candidate dimensions $D \gg 10^3$ vastly exceed sample observations, traditional evolutionary wrappers rely on undirected stochastic bit flips with a uniform mutation probability $p_{\text{mut}} = 1/D$. Under this blind perturbation regime, meta-heuristics inevitably suffer from two structural search lesions. First, when an established, highly informative biomarker with active status $x_{e,j} = 1$ is blindly mutated to 0, delicate multi-feature synergies are dismantled, inducing severe classification degradation. Second, when an uninformative background noise feature with inactive state $x_{e,j} = 0$ is randomly flipped to 1, the candidate subset expands unnecessarily, escalating evaluation latency and precipitating classifier overfitting. Because the probability of randomly hitting decisive dimensions within a 2^D combinatorial neighborhood is asymptotically infinitesimal, scaling as $O(1/D)$, undirected mutation squanders function evaluations on destructive trials. Furthermore, conventional meta-heuristics exhibit severe search amnesia, as historical fitness feedback from intermediate iterations is discarded immediately after particle position updates.

3.3.1. Online feature importance learning

In ultra-high-dimensional wrapper feature selection, standard evolutionary algorithms suffer from severe search amnesia. Although thousands of candidate feature subsets are iteratively synthesized and evaluated via cross-validation, historical fitness differentials produced by coordinate transitions are discarded immediately following generational updates. Consequently, evolutionary trajectories remain memoryless, repeatedly re-evaluating uninformative dimensions and barren combinatorial regions. To bridge generational evaluations with cumulative historical learning, KDS-EL maintains a persistent continuous feature-importance vector $\mathbf{W} = [W_1, \ldots, W_D] \in \mathbb{R}^D$, initialized to the null vector $\mathbf{0}$, which systematically records the directional contribution of each candidate dimension toward classification discriminability.

At generation t, whenever an updated candidate solution $\mathbf{x}_i^{(t)} \in \{0,1\}^D$ achieves a strict fitness improvement over its historical personal best $\mathbf{p}_{\text{best},i} \in \{0,1\}^D$, the algorithm isolates the differential feature set $C_i^{(t)} \subset \{1, \ldots, D\}$ containing coordinates whose binary states were altered during the evolutionary update:

$$C_i^{(t)} = \left\{ j \in \{1, \ldots, D\} \;\middle|\; x_{i,j}^{(t)} \neq p_{\text{best},i,j} \right\}. \tag{15}$$

Let $\delta_i^{(t)} = f(\mathbf{p}_{\text{best},i}, \mathbf{w}(t)) - f(\mathbf{x}_i^{(t)}, \mathbf{w}(t)) > 0$ denote the scalar fitness improvement evaluated under dynamic weight vector $\mathbf{w}(t)$. To distinguish beneficial feature inclusions from advantageous feature eliminations, each altered coordinate $j \in C_i^{(t)}$ receives a directional credit assignment scaled by $\delta_i^{(t)}$:

$$W_j \leftarrow W_j + \delta_i^{(t)} \cdot (2x_{i,j}^{(t)} - 1), \quad \forall j \in C_i^{(t)}, \tag{16}$$

where the polarity factor $(2x_{i,j}^{(t)} - 1) \in \{-1, +1\}$ yields $+1$ when feature j was newly activated ($x_{i,j}^{(t)} = 1$) to achieve the superior solution, reinforcing its importance score, and yields -1 when feature j was successfully pruned ($x_{i,j}^{(t)} = 0$), penalizing its inclusion. Conversely, candidate solutions that fail to improve fitness leave the cumulative importance vector $\mathbf{W}$ unaltered, shielding the learned landscape from noisy perturbations.

Because candidate subsets become progressively sparser across evolution, unconstrained accumulation would allow early-discovered, coarsely predictive dimensions to dominate $\mathbf{W}$ indefinitely. Such historical inertia suppresses emergent synergistic biomarker interactions that only become decisive once redundant dimensions are eliminated. To reconcile long-term knowledge retention with responsiveness to emergent feature synergies, an exponential memory decay is applied at the conclusion of each generation:

$$\mathbf{W} \leftarrow \gamma \cdot \mathbf{W}, \tag{17}$$

where $\gamma \in (0, 1)$ denotes the continuous memory retention factor governing the exponential forgetting rate. Consequently, dimensions that consistently yield fitness gains sustain prominent positive scores, whereas transient or uninformative coordinates undergo geometric attenuation toward zero. Crucially, as $\gamma \to 1$, historical search credit accumulates without forgetting, risking exploratory fixation; conversely, as $\gamma \to 0$, memory collapses into single-step myopic feedback. Maintaining $\gamma \in (0, 1)$ ensures that the importance vector provides a calibrated probabilistic landscape that directly steers discrete sampling in Subpopulation $\mathcal{P}_2$ and guides the directional elite refinement strategy.

3.3.2. Knowledge-guided elite refinement

In ultra-high-dimensional regimes where candidate dimensions $D \gg 10^3$ vastly exceed sample observations, uniform bit-flipping degenerates into blind trial-and-error, disrupting synergistic feature coalitions and precipitating search stagnation. Moreover, applying local search across all particles incurs prohibitive computational latency scaling with swarm size N. To overcome this combinatorial dilemma, the KER strategy exploits the continuous feature-importance landscape $\mathbf{W}$ to steer directional bit inversions strictly on a compact elite echelon $k_e \ll N$, concentrating search momentum on high-potential dimensions while bounding evaluation overhead to exactly k_e fitness calculations per generation.

At each generation, the top k_e individuals ranked by objective fitness are admitted into the elite pool. For each elite candidate $\mathbf{x}_e \in \{0,1\}^D$, the number of perturbed features is set to $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$, where $\phi \in (0, 1)$ represents the perturbation fraction. As depicted in the operational schematic of Fig. 4, the KER strategy probabilistically executes one of two operational pathways:

Branch 1: Ergodic random exploration: To preserve search ergodicity and prevent the elite pool from stagnating in deceptive local basins, a feature index subset $\mathcal{S}_{\text{rand}} \subset \{1, \ldots, D\}$ of cardinality k_{flip} is sampled uniformly at random

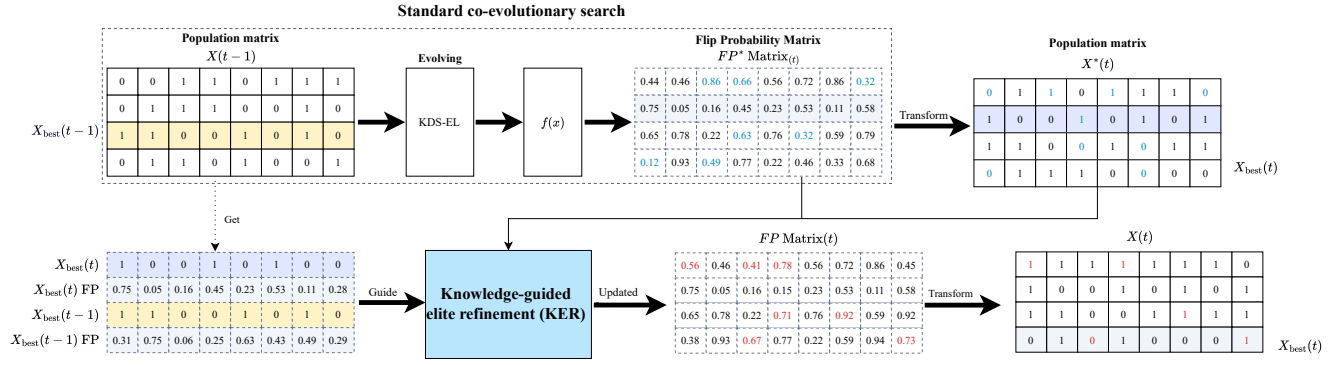


Figure 4: Operational mechanism of the knowledge-guided elite refinement strategy.

Algorithm 1 Knowledge-guided elite refinement

Require: Elite candidate solution $\mathbf{x}_e \in \{0,1\}^D$, cumulative importance vector $\mathbf{W} \in \mathbb{R}^D$, perturbation fraction $\phi \in (0,1)$, exploration probability $p_{\text{rand}} \in [0,1]$, dynamic weight vector $\mathbf{w}(t) \in [0,1]^2$
Ensure: Refined elite candidate solution $\mathbf{x}_e^\star \in \{0,1\}^D$

$k_{\text{flip}} \leftarrow \max(1, \lfloor \phi \cdot D \rfloor)$
$\mathbf{x}'_e \leftarrow \mathbf{x}_e$ ▷ Initialize mutant candidate
Sample random variable $r \sim \mathcal{U}(0,1)$
if $r < p_{\text{rand}}$ **then** ▷ Branch 1: Ergodic random exploration
  Sample random feature index subset $\mathcal{S}_{\text{rand}} \subset \{1,\dots,D\}$ with $|\mathcal{S}_{\text{rand}}| = k_{\text{flip}}$
  $x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}$
else ▷ Branch 2: Knowledge-guided directional refinement
  for $j = 1$ **to** D **do**
    $\sigma_j \leftarrow (1 - 2x_{e,j}) \cdot W_j$ ▷ Compute directional flip score via (19)
  end for
  Identify index subset $\mathcal{S}_{\text{know}}$ of size k_{flip} with top σ_j scores via (20)
  $x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{know}}$
end if
Evaluate candidate fitness $f(\mathbf{x}'_e, \mathbf{w}(t))$ via (23)
if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t))$ **then** ▷ Greedy pairwise in-place replacement
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}'_e$
else
  $\mathbf{x}_e^\star \leftarrow \mathbf{x}_e$
end if
return $\mathbf{x}_e^\star$

with exploration probability $p_{\text{rand}} \in [0,1]$, and the corresponding coordinates are inverted:

$$x'_{e,j} \leftarrow 1 - x_{e,j}, \quad \forall j \in \mathcal{S}_{\text{rand}}, \quad |\mathcal{S}_{\text{rand}}| = k_{\text{flip}}. \tag{18}$$

Branch 2: Knowledge-guided directional refinement: With complementary probability $1-p_{\text{rand}}$, the strategy executes targeted directional refinement driven by the cumulative importance landscape. For each candidate dimension $j \in \{1,\dots,D\}$, a state-dependent directional flip score $\sigma_j \in \mathbb{R}$ is formulated as:

$$\sigma_j = (1 - 2x_{e,j}) \cdot W_j = \begin{cases} +W_j, & \text{if } x_{e,j} = 0, \\ -W_j, & \text{if } x_{e,j} = 1. \end{cases} \tag{19}$$

A positive flip score $\sigma_j > 0$ reflects two favorable search transitions: an unselected feature with inactive coordinate $x_{e,j} = 0$ that exhibits a positive historical contribution $W_j > 0$, or an active feature with $x_{e,j} = 1$ whose elimination has historically improved classification fitness with $W_j < 0$. In either scenario, flipping the binary state of feature j is expected to enhance candidate solution quality. The feature subset $\mathcal{S}_{\text{know}}$ containing the k_{flip} features with the largest directional flip scores is identified:

$$\mathcal{S}_{\text{know}} = \arg \max_{\substack{\mathcal{S} \subseteq \{1,\dots,D\} \\ |\mathcal{S}| = k_{\text{flip}}}} \sum_{j \in \mathcal{S}} \sigma_j, \tag{20}$$

and the corresponding binary states are inverted via $x'_{e,j} \leftarrow 1 - x_{e,j}$ for all $j \in \mathcal{S}_{\text{know}}$. To resolve potential ties among identical flip scores, an infinitesimal stochastic jitter $\zeta_j \sim \mathcal{N}(0, 10^{-12})$ is added to σ_j, ensuring consistent subset identification.

Greedy pairwise in-place replacement: The perturbed candidate $\mathbf{x}'_e$ is evaluated using the objective fitness function $f(\mathbf{x}'_e, \mathbf{w}(t))$. A greedy acceptance rule determines solution admission:

$$\mathbf{x}_e^\star = \begin{cases} \mathbf{x}'_e, & \text{if } f(\mathbf{x}'_e, \mathbf{w}(t)) < f(\mathbf{x}_e, \mathbf{w}(t)), \\ \mathbf{x}_e, & \text{otherwise.} \end{cases} \tag{21}$$

Upon achieving a strict fitness improvement, $\mathbf{x}'_e$ immediately replaces $\mathbf{x}_e$ in the population and updates the global-best solution if $f(\mathbf{x}'_e, \mathbf{w}(t)) < f_{\text{gbest}}$. Because the elite pool cardinality $k_e \ll N$ comprises only a minor fraction of the overall swarm, the KER strategy operates within a strictly bounded computational overhead of exactly k_e fitness evaluations per generation. The complete procedural logic of elite refinement is formalized in Algorithm 1.

Dual-scale search dynamics and dimension-scaling analysis: The perturbation budget $k_{\text{flip}} = \max(1, \lfloor \phi D \rfloor)$ scales with ambient dimensionality D, producing distinct search behaviors across problem scales. In moderate-dimensional regimes where $D \le 2{,}000$, k_{flip} produces small-scale bit flips that execute fine-grained local neighborhood refinement around elite solutions. Conversely, in ultra-high-dimensional regimes where ambient dimensionality expands past $D > 10^4$, the perturbation budget scales proportionally into hundreds of simultaneous coordinate transitions, causing the KER strategy to shift dynamically from localized perturbation into a guided macro-recombination mechanism. Operating at this broader structural scale allows the swarm to evaluate coordinated multi-feature activations and deactivations guided by $\mathbf{W}$, facilitating transitions across distant combinatorial basins. Crucially, the greedy pairwise in-place replacement rule in (21) serves as an intrinsic safeguard against destructive mutations: whenever an extensive macro-flip degrades classification accuracy or inflates feature penalties, the candidate mutant is discarded without modifying the elite pool. While fixed-fraction scaling $\lfloor \phi D \rfloor$ performs consistently across diverse benchmark scales, adaptively modulating the perturbation budget according to the active subset cardinality $d_{\text{selected}} = \|\mathbf{x}_e\|_0$ via $k_{\text{flip}} = \max(1, \lfloor \phi \cdot \|\mathbf{x}_e\|_0 \rfloor)$, or enforcing an upper saturation cap $k_{\max}$, represents an effective adaptive refinement for extreme-dimensional applications.

3.4. Dynamic two-phase weight scheduling

In ultra-high-dimensional wrapper feature selection where candidate dimensions $D \gg 10^3$ vastly exceed sample observations $N_{\text{samp}} \ll 100$, optimization must balance classification error against subset cardinality. Conventional metaheuristics rely on static scalarization, inducing severe search degeneration. A conservative feature penalty fails to overcome the dense dimensionality attractor, squandering evaluation budgets and precipitating classifier overfitting. Conversely, maintaining an elevated penalty throughout evolution suppresses late-stage convergence, prematurely discarding synergistic biomarker coalitions whose retention slightly increases cardinality but is vital for resolving complex decision boundaries.

To resolve this temporal conflict, KDS-EL introduces the dynamic TPS strategy, decoupling the search horizon into two functional phases, as depicted in the operational flowchart of Fig. 2. The total generation budget T is partitioned at a split point parameterized by the phase transition ratio $\kappa \in (0,1)$. The dynamic objective weight vector $\mathbf{w}(t) = [w_a(t), w_f(t)] \in [0,1]^2$, satisfying the normalization condition $w_a(t) + w_f(t) = 1$, is formulated as:

$$\mathbf{w}(t) = \begin{cases} [w_{a,1}, w_{f,1}], & 1 \le t \le \lfloor \kappa T \rfloor, \\ [w_{a,2}, w_{f,2}], & \lfloor \kappa T \rfloor < t \le T, \end{cases} \tag{22}$$

where the weight coefficients satisfy the strict ordering $w_{f,1} > w_{f,2}$ and $w_{a,1} < w_{a,2}$. At generation t, the unified scalar fitness $f(\mathbf{x}, \mathbf{w}(t))$ of a candidate binary

mask $\mathbf{x} \in \{0,1\}^D$ is evaluated as:

$$f(\mathbf{x}, \mathbf{w}(t)) = w_a(t) \cdot (1 - \mathrm{Acc}(\mathbf{x})) + w_f(t) \cdot \left(\frac{\|\mathbf{x}\|_0}{D}\right), \tag{23}$$

where $\mathrm{Acc}(\mathbf{x}) \in [0,1]$ denotes the 5-fold cross-validation accuracy evaluated by a k-NN classifier with neighborhood parameter $k_{\mathrm{nn}} = 3$, and $\|\mathbf{x}\|_0 = \sum_{j=1}^{D} x_j$ quantifies the active feature cardinality.

The two evolutionary regimes execute complementary optimization roles. Throughout Phase 1, spanning $1 \le t \le \lfloor \kappa T \rfloor$, the elevated feature penalty $w_{f,1} = 0.30$ with $w_{a,1} = 0.70$ exerts decisive pressure against non-informative features, rapidly contracting candidate dimensionality to a compact fraction of D. In Phase 2, spanning $\lfloor \kappa T \rfloor < t \le T$, weights shift to $w_{a,2} = 0.95$ and $w_{f,2} = 0.05$ under split ratio $\kappa = 0.70$, reorienting evolutionary pressure toward fine classification refinement. Because objective weights transition abruptly at generation $t = \lfloor \kappa T \rfloor + 1$, candidate fitness values operate on disparate scales. To eliminate scale distortion and preserve search continuity, all personal-best solutions are instantaneously re-evaluated under Phase 2 weights, setting $f_{\mathrm{pbest},i} \leftarrow f(\mathbf{p}_{\mathrm{best},i}, \mathbf{w}_2)$ for all $i \in \{1, \ldots, N\}$, updating $\mathbf{g}_{\mathrm{best}}$, and resetting stagnation counter $t_{\mathrm{stag}} \leftarrow 0$. This boundary calibration safeguards memory integrity, ensuring Phase 2 builds seamlessly upon Phase 1 pruning without numerical distortion.

3.5. Overall algorithmic procedure and complexity analysis

Algorithm 2 synthesizes the overall execution pipeline of KDS-EL into a unified multi-stage optimization framework. The complete workflow seamlessly coordinates offline filter-level relevance extraction, knowledge-informed hybrid seeding, dynamic two-phase objective scheduling with boundary memory calibration, collaborative dual-subpopulation co-evolution, differential importance harvesting, targeted elite refinement, and adaptive stagnation recovery.

Algorithm 2 Overall algorithmic framework of KDS-EL

Require: Training dataset $\mathcal{D} = \{(\mathbf{f}_i, y_i)\}_{i=1}^{N_{\mathrm{samp}}}$, population size $N \in \mathbb{Z}_+$, maximum generations $T \in \mathbb{Z}_+$, binarization threshold $\theta \in (0,1)$, prior seeding ratio $\rho \in (0,1)$, perturbation scale $\sigma \in \mathbb{R}_+$, forgetting factor $\gamma \in (0,1)$, elite pool size $k_e \in \mathbb{Z}_+$, perturbation fraction $\phi \in (0,1)$, random branch probability $p_{\mathrm{rand}} \in [0,1]$, phase transition split ratio $\kappa \in (0,1)$, stagnation threshold $\tau \in \mathbb{Z}_+$

Ensure: Optimal compact feature subset $\mathbf{x}^\star \in \{0,1\}^D$ and optimal fitness $f_{\mathrm{gbest}} \in \mathbb{R}$

Stage 1: Filter-level relevance extraction

Compute normalized mutual information vector $\mathbf{m} \in [0,1]^D$ via (10) and (11)

Stage 2: Knowledge-informed hybrid population seeding

Initialize population $\mathcal{P} = \{\mathbf{x}_i\}_{i=1}^{N}$ with $\lfloor \rho N \rfloor$ Gaussian MI seeds and $(N - \lfloor \rho N \rfloor)$ uniform particles via (12)–(13)

Initialize importance vector $\mathbf{W} \leftarrow \mathbf{0} \in \mathbb{R}^D$, stagnation counter $t_{\mathrm{stag}} \leftarrow 0$, and initial weights $\mathbf{w}(1) = [w_{a,1}, w_{f,1}]$ via (22)

Evaluate initial candidate fitness values $\{f(\mathbf{x}_i, \mathbf{w}(1))\}_{i=1}^{N}$ via (23); initialize personal bests $\{\mathbf{p}_{\mathrm{best},i}\}_{i=1}^{N}$ and global best $\mathbf{g}_{\mathrm{best}}$

Stage 3: Dynamic two-phase co-evolutionary loop

for $t = 1$ **to** T **do**

  Update dynamic objective weight vector $\mathbf{w}(t)$ according to (22)

  if $t = \lfloor \kappa T \rfloor + 1$ **then** re-evaluate $\{\mathbf{p}_{\mathrm{best},i}\}_{i=1}^{N}$ under $\mathbf{w}_2$, update $\mathbf{g}_{\mathrm{best}}$, reset $t_{\mathrm{stag}} \leftarrow 0$ ▷ Boundary calibration

  Update continuous Subpopulation $\mathcal{P}_1$ via continuous PSO with inter-task knowledge transfer [12]

  Update discrete Subpopulation $\mathcal{P}_2$ via discrete probability sampling guided by $\mathbf{W}$

  Evaluate candidate solution fitness $\{f(\mathbf{x}_i, \mathbf{w}(t))\}_{i=1}^{N}$ using 5-fold cross-validation k-NN classifier via (23)

  Update cumulative importance vector $\mathbf{W}$ with improving differentials and apply memory decay γ via (16)–(17)

  Update personal bests $\{\mathbf{p}_{\mathrm{best},i}\}_{i=1}^{N}$ and identify top k_e elite candidate indices $\mathcal{E}$

  for each $e \in \mathcal{E}$ **do** ▷ Knowledge-guided elite refinement

    $\mathbf{p}_{\mathrm{best},e} \leftarrow \mathrm{KER}(\mathbf{p}_{\mathrm{best},e}, \mathbf{W}, \phi, p_{\mathrm{rand}}, \mathbf{w}(t))$ ▷ Invoke Algorithm 1

  end for

  if $\min_i f(\mathbf{p}_{\mathrm{best},i}, \mathbf{w}(t)) < f(\mathbf{g}_{\mathrm{best}}, \mathbf{w}(t))$ **then**

    Update global best $\mathbf{g}_{\mathrm{best}} \leftarrow \arg\min_i f(\mathbf{p}_{\mathrm{best},i}, \mathbf{w}(t))$ and reset $t_{\mathrm{stag}} \leftarrow 0$

  else

    Increment stagnation counter $t_{\mathrm{stag}} \leftarrow t_{\mathrm{stag}} + 1$

  end if

  if $t_{\mathrm{stag}} \ge \tau$ **then** ▷ Adaptive stagnation recovery

    Refresh lower-performing half of population using Gaussian-perturbed MI seeds via (14)

    Re-evaluate refreshed particles and reset $t_{\mathrm{stag}} \leftarrow 0$

  end if

end for

return $\mathbf{x}^\star \leftarrow \mathbf{g}_{\mathrm{best}}$, $f_{\mathrm{gbest}} \leftarrow f(\mathbf{g}_{\mathrm{best}}, \mathbf{w}(T))$

3.5.1. Computational complexity analysis

The computational overhead of KDS-EL decouples into two distinct operational regimes: one-time offline preprocessing and online generational evolution. In the offline phase, evaluating non-parametric mutual information scores across D candidate dimensions with N_{samp} training samples via k-nearest neighbors search requires $O(D \cdot N_{\mathrm{samp}} \log N_{\mathrm{samp}})$ operations, while normalizing and sorting these relevance estimates incurs $O(D \log D)$ time. Because this procedure is executed exactly once prior to evolutionary search, the offline preprocessing complexity $\mathcal{T}_{\mathrm{pre}} = O(D \cdot N_{\mathrm{samp}} \log N_{\mathrm{samp}} + D \log D)$ represents a negligible fraction of the total budget, particularly in small-sample regimes where $N_{\mathrm{samp}} \ll 100$. Throughout online evolution, each generation $t \in \{1, \ldots, T\}$ orchestrates three coordinated operational stages. First, subpopulation updates and cross-task knowledge transfer—comprising continuous velocity-position updates in $\mathcal{P}_1$ and discrete probabilistic sampling in $\mathcal{P}_2$—require $O(N \cdot D)$ arithmetic operations per generation. Second, wrapper fitness evaluation constitutes the dominant computational expenditure: for each candidate solution $\mathbf{x}_i$ with active feature cardinality $d_i = \|\mathbf{x}_i\|_0$, computing pairwise Euclidean distances under 5-fold cross-validation with a k-NN classifier incurs $O(N_{\mathrm{samp}}^2 \cdot d_i)$ runtime, aggregating to $O(N \cdot N_{\mathrm{samp}}^2 \cdot \bar{d}^{(t)})$ across the swarm of size N, where $\bar{d}^{(t)} = \frac{1}{N}\sum_{i=1}^{N} d_i$ denotes the average active feature count at generation t. Third, cumulative importance learning and elite refinement introduce targeted search intensification within strictly bounded overhead: extracting differential feature masks and updating memory vector $\mathbf{W}$ consumes $O(N \cdot D)$ operations; computing directional flip scores across all D dimensions and isolating top k_{flip} features incurs $O(D \log D)$ time via partial quick-sort; and re-evaluating refined elite candidates requires $O(k_e \cdot N_{\mathrm{samp}}^2 \cdot d_{\mathrm{elite}})$ operations. Because elite pool size is strictly bounded with $k_e \ll N$, where $k_e = 3$, this refinement cost represents a marginal fraction of generational overhead. Accumulating these operational costs across T generations yields the total asymptotic time complexity of KDS-EL:

$$\mathcal{T}_{\mathrm{total}} = O\left(T \cdot N \cdot N_{\mathrm{samp}}^2 \cdot \bar{d}_{\mathrm{selected}} + T \cdot N \cdot D\right), \tag{24}$$

where $\bar{d}_{\mathrm{selected}}$ denotes the average active feature count across all evolutionary generations.

Although KDS-EL shares the worst-case asymptotic upper bound $O(T \cdot N \cdot N_{\mathrm{samp}}^2 \cdot D)$ with standard swarm feature selectors, its practical runtime is substantially smaller. This computational advantage stems directly from the cooperative interplay between the prior-guided sparse population initialization strategy and the dynamic two-phase weight scheduling strategy. Rather than evaluating dense candidate subsets collapsing into the unguided dimensionality attractor $d \approx D/2$, KDS-EL aggressively contracts active feature cardinality to an ultra-sparse subset $\bar{d}_{\mathrm{selected}} \ll D$ from the earliest generations. Because pairwise Euclidean distance evaluations in the k-NN classifier scale strictly linearly with selected features d_{selected}, reducing active dimensionality by orders of magnitude dramatically curtails arithmetic operations per evaluation, converting theoretical sparsity into substantial practical runtime acceleration.

Regarding spatial footprint, memory consumption comprises the binary population $\mathbf{X} \in \{0,1\}^{N \times D}$, continuous positions $\mathbf{P} \in [0,1]^{N \times D}$, velocities $\mathbf{V} \in \mathbb{R}^{\lfloor N/2 \rfloor \times D}$, personal bests $\mathbf{P}_{\mathrm{best}} \in \{0,1\}^{N \times D}$, feature importance memory $\mathbf{W} \in \mathbb{R}^D$, and training data $\mathcal{D} \in \mathbb{R}^{N_{\mathrm{samp}} \times D}$. The resulting total spatial complexity is $\mathcal{S}_{\mathrm{total}} = O\left((3N + N_{\mathrm{samp}} + 1)D\right) = O\left((N + N_{\mathrm{samp}})D\right)$, scaling strictly linearly with feature dimensionality D. Because allocations use standard contiguous arrays, working memory remains on the order of megabytes even for ultra-high-dimensional datasets with $D > 10^5$, ensuring seamless execution on commodity hardware without memory overhead.

4. Experimental results and discussion

This section outlines the experimental framework and presents a comprehensive empirical evaluation of KDS-EL across eight analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects. All aggregate numbers are computed over the 20 benchmark datasets described in Section 4.1.

4.1. Experimental setup

This subsection outlines the experimental framework designed to evaluate the classification performance, dimensionality reduction capability, computational efficiency, and hyperparameter robustness of KDS-EL.

4.1.1. High-dimensional benchmark datasets

To validate the performance and generalization ability of KDS-EL across diverse problem domains, a total of 20 benchmark datasets widely adopted in the evolutionary feature selection literature [8, 11, 12] are selected. These datasets encompass diverse categories and sample sizes across biomedical gene-expression microarrays, text categorization corpora, image recognition, speech processing, and synthetic benchmarks. Such diverse data distributions and dimensional complexities effectively evaluate the generalization capability and feature reduction efficacy of feature selection algorithms under challenging conditions. The detailed characteristics of the investigated datasets are summarized in Table 2.

4.1.2. Comparison algorithms

We benchmark KDS-EL against nine representative single-objective competitor algorithms across two primary categories, as categorized in Table 1. The first category comprises established swarm methods, including FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7]. The second category encompasses contemporary metaheuristics, including the original MEL algorithm [12], DENCA-SO [14], bIRIME [13], and BinCOA [15], configured under parameters in Table 3. These competitors systematically evaluate single-objective scalarized wrapper fitness under identical evaluation budgets, guaranteeing empirical parity. Contemporaneous knowledge-guided metaheuristics, such as NNS-DFPG-CSA [17] and MI-MPODE [34], pursue multi-objective Pareto fronts with external archives, representing an orthogonal optimization paradigm to the scalarized wrapper framework investigated here.

4.1.3. Evaluation metrics and statistical test protocols

Algorithm performance is evaluated across three primary quantitative criteria: classification accuracy with standard deviation, the number of selected features alongside the feature selection ratio, and computational runtime in seconds. A 3-nearest neighbor (3-NN) classifier with 5-fold stratified cross-validation serves as the wrapper evaluation model across all benchmark datasets, ensuring parameter-free reproducibility and high sensitivity to candidate feature subset discriminability. Following the standard benchmarking protocol established in high-dimensional evolutionary feature selection literature [8, 11, 12], candidate subsets are evaluated using stratified cross-validation as the internal fitness surrogate. Under this standardized formulation, each benchmark problem is treated as a self-contained combinatorial optimization instance where cross-validation accuracy guides evolutionary search trajectories under identical computational budgets across all competing methods.

To guarantee experimental fairness and statistical reliability, all algorithms are evaluated over 10 independent runs with distinct random seeds under the standardized parameter settings detailed in Table 3. In ultra-high-dimensional wrapper feature selection, each candidate evaluation requires repeated classifier validation across thousands or tens of thousands of candidate dimensions, reaching up to 54,675 dimensions on *T1D*, thereby imposing an intensive computational burden. Conducting 10 independent runs corresponds to 50 distinct stratified cross-validation folds per benchmark dataset, providing an adequate sample size to capture classifier generalization variance and algorithmic stability while remaining computationally tractable across the extensive comparison suite. This evaluation protocol strictly mirrors the benchmark standard established in top-tier literature on ultra-high-dimensional feature selection, particularly the original MEL algorithm published in *IEEE Transactions on Knowledge and Data Engineering* [12].

Furthermore, to determine whether the empirical advantages of KDS-EL over competing paradigms are statistically significant rather than stochastic artifacts, we strictly adhere to the multi-problem non-parametric statistical testing framework recommended by Demšar [39]. In this protocol, statistical power is governed by the 20 independent benchmark datasets, with each problem represented by its mean classification accuracy across the 10 independent runs. Specifically, the non-parametric Friedman test [40] is utilized to establish global omnibus algorithm rankings across the benchmark suite. Pairwise Wilcoxon signed-rank tests at the $\alpha = 0.05$ significance level assess directional superiority between KDS-EL and each competitor algorithm, with the Holm–Bonferroni step-down procedure [41] applied to rigorously control the family-wise error rate across the multiple competitor comparisons.

4.2. Experimental analysis

This subsection presents a comprehensive empirical evaluation of KDS-EL across eight complementary analytical dimensions: classification accuracy, feature subset compactness, fitness convergence dynamics, statistical significance, performance progression before and after algorithmic improvement, computational efficiency, hyperparameter sensitivity, and in-depth discussion on biomedical ceiling effects.

Table 2: Characteristics of the 20 benchmark datasets.

No.	Dataset	Samples (n)	Features (D)	Classes (C)	Domain
1	Adenoma	36	7,457	2	Gene expression
2	ALL_AML	72	7,129	2	Gene expression
3	ALL1	128	12,625	2	Gene expression
4	ALL4	93	12,625	2	Gene expression
5	Colon	62	2,000	2	Gene expression
6	DLBCL	77	7,129	2	Gene expression
7	Gastric	65	22,645	2	Gene expression
8	Gastric1	144	22,283	2	Gene expression
9	Gastric2	124	22,283	2	Gene expression
10	Leukaemia	72	7,129	2	Gene expression
11	Lymphoma	45	4,026	2	Gene expression
12	Myeloma	173	12,625	2	Gene expression
13	Prostate	102	12,625	2	Gene expression
14	SRBCT	83	2,308	4	Gene expression
15	T1D	101	54,675	2	Gene expression
16	BASEHOCK	1,993	4,862	2	Text
17	PCMAC	1,943	3,289	2	Text
18	COIL20	1,440	1,024	20	Image
19	ISOLET	1,560	617	26	Speech
20	madelon	2,600	500	2	Synthetic

Table 3: Relevant parameter settings for all algorithms.

Algorithm / Component	Parameter setting
Common experimental parameters:	
Population size N	20
Maximum generations $T_{\max}$	100
Classifier	3-NN with 5-fold stratified CV
Binarization threshold θ	0.60
Independent runs	10
Proposed KDS-EL framework:	
Subpopulations	$N_1 = 10,\ N_2 = 10$
Inertia weight ω	0.90
Acceleration coefficients c_1, c_2, c_3	2.0, 2.0, 2.0
Knowledge decay factor γ	0.99
Perturbation fraction ϕ	0.01
Elite pool size k_e	3
Random exploration rate p_{rand}	0.10
MI initialization ratio ρ	0.50
Stagnation threshold τ	20 generations
Phase split ratio κ	0.70
Phase-1 weights $\mathbf{w}_1$	[0.70, 0.30]
Phase-2 weights $\mathbf{w}_2$	[0.95, 0.05]
Intermediate configurations:	
KDS-EL w/o TPS	Static weights $\mathbf{w}_s$ = [0.90, 0.10]
KDS-EL w/o TPS&KER	Without KER and TPS, $\mathbf{w}_s$ = [0.90, 0.10]
Original MEL [12]	Uniform init, $\mathbf{w}_s$ = [0.90, 0.10]
Competitor algorithms:	
FWPSO [8]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ \alpha_{\text{sa}} = 0.95$
PSO-CSM [11]	$\omega = 0.7298,\ c_1 = c_2 = 1.49618,\ p_m = 0.05$
SaWDE [9]	$F \in [0.1, 0.9],\ CR \in [0.1, 0.9]$
IGPSO [10]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0$, IG scoring
BPSO [7]	$\omega \in [0.4, 0.9],\ c_1 = c_2 = 2.0,\ V_{\max} = 6.0$
DENCA-SO [14]	$F = 0.5,\ CR = 0.2,\ W = 0.05$, NCFS $\lambda = 0.01$
bIRIME [13]	S-transfer, CMS mutation, restart limit $L = 5$
BinCOA [15]	ROBL, Sigmoid transfer, crisscross (HC/VC)

4.2.1. Comparison of classification accuracy

The KDS-EL framework was systematically evaluated alongside nine competitor algorithms and two intermediate algorithmic configurations across diverse problem domains. Table 4 summarizes the classification accuracy results across 10 independent runs, with the best values in boldface and the

Table 4: Comparison of classification accuracy between KDS-EL and competitor algorithms.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	0.9998	0.9996 $^{+}$	0.9996 $^{+}$	0.9664 $^{+}$	**1.0000** $^{-}$	0.9952 $^{+}$	0.9900 $^{+}$	**1.0000** $^{-}$	0.9710 $^{+}$	0.9802 $^{+}$	**1.0000** $^{-}$	0.9534 $^{+}$
ALL4	**0.9522**	0.9418 $^{\approx}$	0.9415 $^{+}$	0.8704 $^{+}$	0.9140 $^{+}$	0.8617 $^{+}$	0.8363 $^{+}$	0.7634 $^{+}$	0.8062 $^{+}$	0.7914 $^{+}$	0.7043 $^{+}$	0.7569 $^{+}$
ALL_AML	0.9995	0.9990 $^{+}$	0.9886 $^{+}$	0.9539 $^{+}$	**1.0000** $^{-}$	0.9850 $^{+}$	0.9748 $^{+}$	0.9861 $^{+}$	0.9591 $^{+}$	0.9669 $^{+}$	0.9778 $^{+}$	0.9426 $^{+}$
Adenoma	0.9996	0.9996 $^{\approx}$	0.9958 $^{+}$	0.9410 $^{+}$	**1.0000** $^{-}$	0.9783 $^{+}$	0.9665 $^{+}$	0.9722 $^{+}$	0.9461 $^{+}$	0.9527 $^{+}$	0.9611 $^{+}$	0.9287 $^{+}$
BASEHOCK	**0.9113**	0.8997 $^{+}$	0.8597 $^{+}$	0.8518 $^{+}$	0.9082 $^{\approx}$	0.8715 $^{+}$	0.8637 $^{+}$	0.8897 $^{+}$	0.8860 $^{+}$	0.8479 $^{+}$	0.8610 $^{+}$	0.8597 $^{+}$
COIL20	0.9939	0.9919 $^{+}$	0.9619 $^{+}$	0.9634 $^{+}$	**0.9991** $^{-}$	0.9894 $^{+}$	0.9782 $^{+}$	0.9953 $^{-}$	0.9763 $^{+}$	0.9707 $^{+}$	0.9942 $^{\approx}$	0.9573 $^{+}$
Colon	**0.9500**	0.9347 $^{+}$	0.9211 $^{+}$	0.8548 $^{+}$	0.9468 $^{\approx}$	0.9221 $^{+}$	0.8880 $^{+}$	0.8694 $^{+}$	0.8867 $^{+}$	0.8636 $^{+}$	0.8048 $^{+}$	0.8529 $^{+}$
DLBCL	0.9995	0.9993 $^{\approx}$	0.9902 $^{+}$	0.9483 $^{+}$	**1.0000** $^{-}$	0.9819 $^{+}$	0.9819 $^{+}$	0.9883 $^{+}$	0.9629 $^{+}$	0.9655 $^{+}$	0.9675 $^{+}$	0.9435 $^{+}$
Gastric	**0.9782**	0.9677 $^{+}$	0.9748 $^{\approx}$	0.8955 $^{+}$	0.9569 $^{+}$	0.9191 $^{+}$	0.8874 $^{+}$	0.8231 $^{+}$	0.8234 $^{+}$	0.8564 $^{+}$	0.7769 $^{+}$	0.8046 $^{+}$
Gastric1	**0.9800**	0.9738 $^{+}$	0.9642 $^{+}$	0.9310 $^{+}$	0.9722 $^{+}$	0.9597 $^{+}$	0.9496 $^{+}$	0.9583 $^{+}$	0.9348 $^{+}$	0.9402 $^{+}$	0.9500 $^{+}$	0.9145 $^{+}$
Gastric2	**0.9981**	0.9936 $^{+}$	0.9872 $^{+}$	0.9522 $^{+}$	0.9952 $^{\approx}$	0.9837 $^{+}$	0.9710 $^{+}$	0.9823 $^{+}$	0.9532 $^{+}$	0.9623 $^{+}$	0.9750 $^{+}$	0.9357 $^{+}$
ISOLET	0.8894	0.8898 $^{\approx}$	0.8557 $^{+}$	0.8501 $^{+}$	**0.9112** $^{-}$	0.8822 $^{\approx}$	0.8584 $^{+}$	0.8763 $^{+}$	0.8903 $^{\approx}$	0.8468 $^{+}$	0.8760 $^{+}$	0.8607 $^{+}$
Leukaemia	0.9995	0.9993 $^{\approx}$	0.9895 $^{+}$	0.9534 $^{+}$	**1.0000** $^{-}$	0.9886 $^{+}$	0.9775 $^{+}$	0.9861 $^{+}$	0.9587 $^{+}$	0.9682 $^{+}$	0.9764 $^{+}$	0.9416 $^{+}$
Lymphoma	**0.9994**	**0.9994** $^{\approx}$	0.9868 $^{+}$	0.8731 $^{+}$	0.9978 $^{\approx}$	0.9404 $^{+}$	0.9206 $^{+}$	0.8889 $^{+}$	0.8841 $^{+}$	0.8861 $^{+}$	0.8400 $^{+}$	0.8585 $^{+}$
Myeloma	**0.9155**	0.8971 $^{+}$	0.8825 $^{+}$	0.8450 $^{+}$	0.8884 $^{+}$	0.8702 $^{+}$	0.8550 $^{+}$	0.8387 $^{+}$	0.8522 $^{+}$	0.8304 $^{+}$	0.8012 $^{+}$	0.8186 $^{+}$
PCMAC	**0.8705**	0.8424 $^{+}$	0.8007 $^{+}$	0.7870 $^{+}$	0.8410 $^{+}$	0.8103 $^{+}$	0.8036 $^{+}$	0.8161 $^{+}$	0.8256 $^{+}$	0.7840 $^{+}$	0.7601 $^{+}$	0.7982 $^{+}$
Prostate	**0.9701**	0.9595 $^{+}$	0.9638 $^{+}$	0.8458 $^{+}$	0.9353 $^{+}$	0.8887 $^{+}$	0.8721 $^{+}$	0.8686 $^{+}$	0.8711 $^{+}$	0.8526 $^{+}$	0.8235 $^{+}$	0.8485 $^{+}$
SRBCT	0.9986	0.9984 $^{\approx}$	0.9944 $^{+}$	0.9321 $^{+}$	**1.0000** $^{-}$	0.9692 $^{+}$	0.9695 $^{+}$	0.9663 $^{+}$	0.9659 $^{+}$	0.9515 $^{+}$	0.9325 $^{+}$	0.9488 $^{+}$
T1D	**0.8598**	0.8232 $^{+}$	0.8246 $^{+}$	0.7679 $^{+}$	0.7752 $^{+}$	0.7610 $^{+}$	0.7318 $^{+}$	0.7178 $^{+}$	0.7237 $^{+}$	0.7113 $^{+}$	0.6723 $^{+}$	0.7063 $^{+}$
madelon	**0.8822**	0.8659 $^{\approx}$	0.7679 $^{+}$	0.7521 $^{+}$	0.8677 $^{\approx}$	0.7963 $^{+}$	0.7669 $^{+}$	0.7598 $^{+}$	0.8106 $^{+}$	0.7497 $^{+}$	0.7695 $^{+}$	0.7652 $^{+}$
Average	0.9574	0.9488	0.9325	0.8868	0.9454	0.9177	0.9021	0.8973	0.8944	0.8839	0.8712	0.8698
$+/\approx/-$	—	12/8/0	19/1/0	20/0/0	7/5/8	19/1/0	20/0/0	18/0/2	19/1/0	20/0/0	18/1/1	20/0/0

Entries in bold denote the best classification accuracy, while underlined entries indicate the second-best. $+/\approx/-$ indicates that KDS-EL is statistically significantly superior (+), statistically equivalent ($\approx$), or inferior ($-$) to the corresponding competitor according to the pairwise Wilcoxon signed-rank test at $\alpha = 0.05$.

second-best underlined.

Across the benchmark suite, KDS-EL achieves the highest overall mean classification accuracy of 0.9574, securing decisive predictive leadership in multi-dataset aggregate performance. On ultra-high-dimensional microarrays characterized by severe sample sparsity, such as *ALL4*, *Colon*, and *T1D*, KDS-EL establishes pronounced discriminative advantages over peer methods. Compared with established single-objective evolutionary algorithms, including FWPSO [8], PSO-CSM [11], SaWDE [9], IGPSO [10], and BPSO [7], KDS-EL substantially elevates classification performance, confirming that closed-loop knowledge transfer and dynamic weight scheduling effectively mitigate random exploratory drift. Furthermore, against contemporary competitor methods published between 2023 and 2024, including BinCOA [15] and DENCA-SO [14], KDS-EL demonstrates consistent predictive advantages. Compared with the original MEL algorithm [12], KDS-EL achieves an accuracy improvement exceeding seven percentage points, demonstrating that prior-guided initialization and knowledge-guided elite refinement fundamentally enhance localized exploitation in complex biomarker landscapes.

A granular examination of per-dataset performance in Table 4 reveals nuanced trade-off dynamics between KDS-EL and the strongest contemporary competitor, bIRIME [13]. Head-to-head on classification accuracy alone, KDS-EL records 7 wins, 5 ties, and 8 losses against bIRIME across the 20 benchmark datasets. Crucially, six of these eight losses occur strictly on saturated ceiling-effect microarrays—including *ALL1*, *ALL_AML*, *Adenoma*, *DLBCL*, *Leukaemia*, and *SRBCT*—where classification accuracy for both algorithms exceeds 0.998 and variations remain below 0.0015, reflecting negligible stochastic fluctuations rather than genuine predictive divergence. Conversely, on challenging, non-saturated problem domains, KDS-EL establishes decisive margins over bIRIME, securing substantial accuracy improvements of 8.46 pp on *T1D*, 3.82 pp on *ALL4*, and 3.48 pp on *Prostate*. Furthermore, as documented in Table 5, bIRIME secures its marginal ceiling gains at the expense of severe feature bloat, retaining an average of 424.8 features across the benchmark suite—nearly five times the 87.8 features isolated by KDS-EL. On *DLBCL*, for instance, bIRIME attains a perfect accuracy of 1.0000 by preserving 247 features, whereas KDS-EL achieves a statistically equivalent accuracy of 0.9995 with merely 64 features, eliminating nearly three-quarters of redundant dimensions. Additionally, as analyzed in Section 4.2.6, bIRIME incurs a steep wall-clock runtime penalty averaging 375.2 seconds per instance due to unguided multi-operator evaluations, running more than seven times slower than KDS-EL. Consequently, KDS-EL achieves a superior omnibus mean accuracy of 0.9574 compared with 0.9454 for bIRIME while delivering unprecedented dimensional parsimony, stability, and computational thrift.

4.2.2. *Comparison of the number of selected features*

Table 5 summarizes the selected feature subset size for each problem instance across the evaluated competitor algorithms and intermediate configurations, with the most compact subset sizes highlighted in boldface. Across the benchmark suite, KDS-EL achieves unprecedented feature subset compactness, identifying an overall average signature of merely 87.8 features and compressing ultra-high-dimensional feature spaces by over 98%. This dimensionality reduction capability is particularly pronounced on complex biomedical microarrays, such as *ALL1*, *Colon*, *Gastric1*, and *Prostate*, where KDS-EL isolates minimal biomarker subsets of fewer than 40 genes while maintaining near-perfect classification performance.

In contrast, conventional evolutionary feature selection algorithms exhibit severe dimensional redundancy. Even the most competitive peer wrapper, FWPSO [8], retains an average of over 1,000 features, while contemporary metaheuristics such as bIRIME [13] preserve over 400 features due to the absence of targeted importance decay guidance. Most critically, canonical swarm heuristics, including BinCOA [15], DENCA-SO [14], and standard BPSO [7], remain fundamentally trapped in the uninformative 50% dimensionality attractor, selecting over 5,000 redundant features across high-dimensional landscapes. This severe degradation occurs because unguided binarization mechanisms, such as standard sigmoid or uniform thresholding, inherently center active dimension probabilities around the mathematical expectation of $D/2$ when proactive pruning pressure is absent. By contrast, KDS-EL combines continuous mutual information prior seeding with aggressive early-stage feature penalization in Two-Phase Weight Scheduling, actively driving candidate trajectories away from the $D/2$ attractor into ultra-sparse, discriminative subspaces.

Evaluating the intermediate algorithmic configurations demonstrates how each constituent mechanism systematically compresses feature redundancy. Across the benchmark suite, feature cardinality contracts hierarchically from over 2,700 dimensions in the original MEL algorithm [12] down to fewer than 300 dimensions through mutual information prior seeding, further halves via knowledge-guided elite refinement, and ultimately consolidates into an ultra-compact signature of 87.8 features under dynamic two-phase scheduling. This progressive reduction confirms that the proposed components operate in a complementary cascade, where domain prior seeding eliminates global noise, elite refinement purges weakly contributing dimensions, and two-phase scheduling enforces rigorous dimensional parsimony.

Crucially, while feature reduction progresses monotonically in terms of the multi-dataset average, subset cardinality on individual problem instances reflects a deliberate, domain-specific trade-off governed by two-phase scheduling dynamics. On several challenging biomedical microarrays, such as *Adenoma*, *Lymphoma*, and *SRBCT*, the complete KDS-EL framework retains slightly more features than the static-weight variant without TPS. This behavior directly manifests the underlying design rationale of the two-phase strategy: upon transitioning to Phase 2, the feature-cardinality penalty is relaxed while classification accuracy is prioritized. Rather than enforcing an inflexible sparsity constraint that risks permanently pruning decisive biomarker interactions, this controlled relaxation empowers the search engine to strategically retrieve a small subset of weakly individual but strongly synergistic features that resolve ambiguous classification boundaries. Consequently, this targeted trade-off converts marginal feature retention into a statistically confirmed accuracy advantage, confirming that KDS-EL successfully harmonizes extreme feature

Table 5: Comparison of selected feature subset size between KDS-EL and competitor algorithms.

Dataset	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
ALL1	**39**	55	55	4248	65	612	1206	6120	3667	2501	5877	5886
ALL4	124	152	**64**	1082	393	1326	1447	6301	4581	2677	6111	6217
ALL_AML	**67**	70	473	2369	119	521	949	3419	1996	1444	3251	3266
Adenoma	66	**33**	120	2483	40	278	937	3554	2102	1608	3397	3399
BASEHOCK	**134**	208	812	1965	1721	2066	1113	2445	1912	1894	2359	2419
COIL20	84	60	348	323	**54**	90	167	454	232	255	634	407
Colon	**14**	35	105	702	113	240	284	960	621	448	845	935
DLBCL	64	**52**	615	2784	247	876	1074	3514	2155	1718	3784	3379
Gastric	100	97	**62**	3604	258	1070	2782	11157	7072	4916	10625	11004
Gastric1	**31**	217	678	7393	213	1420	2966	10937	6963	5074	10911	10661
Gastric2	**83**	140	444	7084	348	1362	3062	11041	6708	5134	10971	10627
ISOLET	**60**	100	225	250	133	172	194	298	213	195	374	272
Leukaemia	66	**47**	488	2409	132	534	934	3406	2026	1440	3218	3246
Lymphoma	45	**23**	50	1404	137	304	524	1934	1204	880	1765	1832
Myeloma	250	365	**175**	2137	1160	1863	1847	6278	4543	2859	6596	6259
PCMAC	**91**	329	637	1339	928	1209	651	1621	1292	1137	1590	1635
Prostate	26	**23**	27	5058	418	1898	1541	6282	4229	2729	5837	6211
SRBCT	64	**36**	129	906	172	431	366	1136	739	515	1056	1083
T1D	330	366	**174**	6715	1459	4984	9047	27318	18009	13170	27145	26971
madelon	**18**	26	130	190	46	122	95	240	162	171	122	221
Average	**87.8**	121.7	290.5	2722.2	407.7	1068.9	1559.4	5420.7	3521.1	2538.2	5323.4	5296.5
Comp. Ratio	**98.0%**	97.2%	91.2%	68.4%	92.7%	86.0%	84.5%	51.3%	68.2%	75.3%	52.3%	53.0%

Entries in bold denote the most compact feature subset size (or highest compression ratio), while underlined entries indicate the second-most compact subset size. Comp. Ratio denotes the average feature reduction percentage relative to the full feature space.

sparsity with optimal predictive discrimination.

4.2.3. Fitness value and convergence analysis

The scalar fitness function (23) formalizes the joint optimization trade-off between minimizing classification error and compressing the active feature ratio, directly quantifying each algorithm's exploratory efficiency and exploitation depth throughout search. To evaluate evolutionary trajectories comprehensively and avoid selective reporting, Fig. 5 plots the mean fitness convergence curves across the benchmark suite over 100 generations for KDS-EL alongside four representative single-objective wrapper methods: FWPSO [8], SaWDE [9], the original MEL algorithm [12], and classical BPSO [7].

As demonstrated across the individual panels in Fig. 5, KDS-EL establishes a decisive convergence advantage, achieving the lowest final fitness values across virtually all evaluated datasets. In the initial generation, mutual information continuous prior seeding directly places the swarm within compact, discriminative subspaces. Consequently, KDS-EL commences search from an initial reference fitness substantially lower than those of competing swarms, which incur severe initial penalties due to unguided random bit assignments that indiscriminately activate thousands of noisy features. During early iterations, the aggressive feature-size penalization in Two-Phase Weight Scheduling accelerates dimensional pruning, producing a steep, monotonic fitness descent across both biomedical microarrays, including *ALL4*, *Colon*, and *Gastric*, and heterogeneous vision and text benchmarks, such as *BASEHOCK* and *COIL20*, while competing methods struggle to escape dense initial configurations.

The comparative trajectories further reveal distinct search stagnation behaviors among the competing paradigms. Standard BPSO [7] exhibits erratic, sluggish progress across high-dimensional landscapes, frequently stagnating after generation 20 due to uniform velocity clipping and the absence of domain guidance. While FWPSO [8] and SaWDE [9] achieve progressive fitness improvements, their trajectories plateau prematurely between generations 40 and 60 because memoryless stochastic perturbations discard iteration feedback and fail to reinforce synergistic feature combinations. Similarly, the original MEL algorithm [12] suffers from an extended exploratory plateau during early generations before locating viable descent directions. In contrast, KDS-EL sustains continuous fitness reduction throughout the entire 100-generation timeline. By dynamically accumulating historical feature-importance feedback within the importance vector $\mathbf{W}$ with exponential decay γ and steering elite perturbations toward high-utility candidates via the KER strategy, KDS-EL continually escapes local attractors, securing an order-of-magnitude fitness reduction on ultra-high-dimensional benchmarks such as *ALL1*, *DLBCL*, *Leukaemia*, *Lymphoma*, and *T1D*.

4.2.4. Statistical significance analysis via Wilcoxon test

To evaluate whether the empirical performance advantages of KDS-EL are statistically meaningful rather than artifacts of stochastic fluctuation, we follow the multi-problem non-parametric statistical testing framework established by Demšar [39]. Table 6 presents the pairwise Wilcoxon signed-rank test outcomes across the benchmark suite at a significance level of $\alpha = 0.05$. Across the comparison suite, KDS-EL demonstrates decisive statistical superiority over established single-objective metaheuristics—such as FWPSO [8], SaWDE [9], PSO-CSM [11], IGPSO [10], and standard BPSO [7]—as well as contemporary competitors including BinCOA [15] and DENCA-SO [14]. Notably, against the original MEL algorithm [12], KDS-EL achieves unanimous wins without conceding a single tie or loss. Furthermore, the statistically confirmed advantage over the intermediate configuration without TPS verifies that dynamic objective scheduling delivers significant predictive gains over static weighting across diverse data landscapes.

The comprehensive pairwise comparison matrix in Table 6 further delineates the competitive structure among the evaluated evolutionary paradigms. Against bIRIME, the pairwise test indicates 7 wins, 5 ties, and 8 losses across the 20 benchmark datasets, reflecting competitive accuracy dynamics between the two top-performing algorithms. However, when evaluated under the scalar fitness objective that simultaneously penalizes feature redundancy, or under multi-criteria evaluation encompassing dimensional parsimony and execution latency, KDS-EL establishes unambiguous superiority. Across the remaining competitor methods, KDS-EL systematically outperforms peer wrapper and filter paradigms, with statistical ties occurring predominantly on saturated ceiling-effect datasets where classification accuracy asymptotically approaches unity, under which KDS-EL matches peer predictive accuracy while isolating far sparser biomarker subsets.

To rigorously guard against family-wise error rate inflation inherent in multiple simultaneous comparisons, the Holm–Bonferroni step-down procedure [41] is conducted with KDS-EL serving as the control method, as detailed in Table 7. Under the nominal family-wise significance level $\alpha = 0.05$, the computed asymptotic p-values across all pairwise comparisons remain strictly below their respective Holm-adjusted critical thresholds. Consequently, all null hypotheses asserting equivalent classification performance between KDS-EL and the comparison methods are formally rejected. This multi-hypothesis verification confirms that the empirical superiority of KDS-EL is statistically robust and reproducible under rigorous multi-dataset benchmarking standards.

To establish an omnibus global algorithmic ranking across the benchmark suite, the non-parametric Friedman test [40] is conducted, with the resulting average ranks illustrated in Fig. 6. KDS-EL secures the top average Friedman rank across the entire comparison suite, outperforming leading peer algorithms such as bIRIME as well as its own intermediate configurations. In contrast, standard swarm algorithms and unguided wrappers occupy noticeably inferior positions, reflecting their vulnerability to premature stagnation in ultra-high dimensions. The Friedman ranking trajectory provides compelling visual and statistical confirmation of KDS-EL's omnibus superiority and robustness across heterogeneous problem distributions.

4.2.5. Performance comparison before and after algorithm improvement

To evaluate constituent mechanism contributions without confounding algorithmic variables, comparative experiments assess a controlled four-stage progression from the original MEL algorithm to KDS-EL. Table 8 summarizes end-to-end performance between both frameworks, with best classification accuracies in boldface and most compact subsets underlined. Complete granular metrics across all intermediate configurations are reported in Tables 4, 5, and

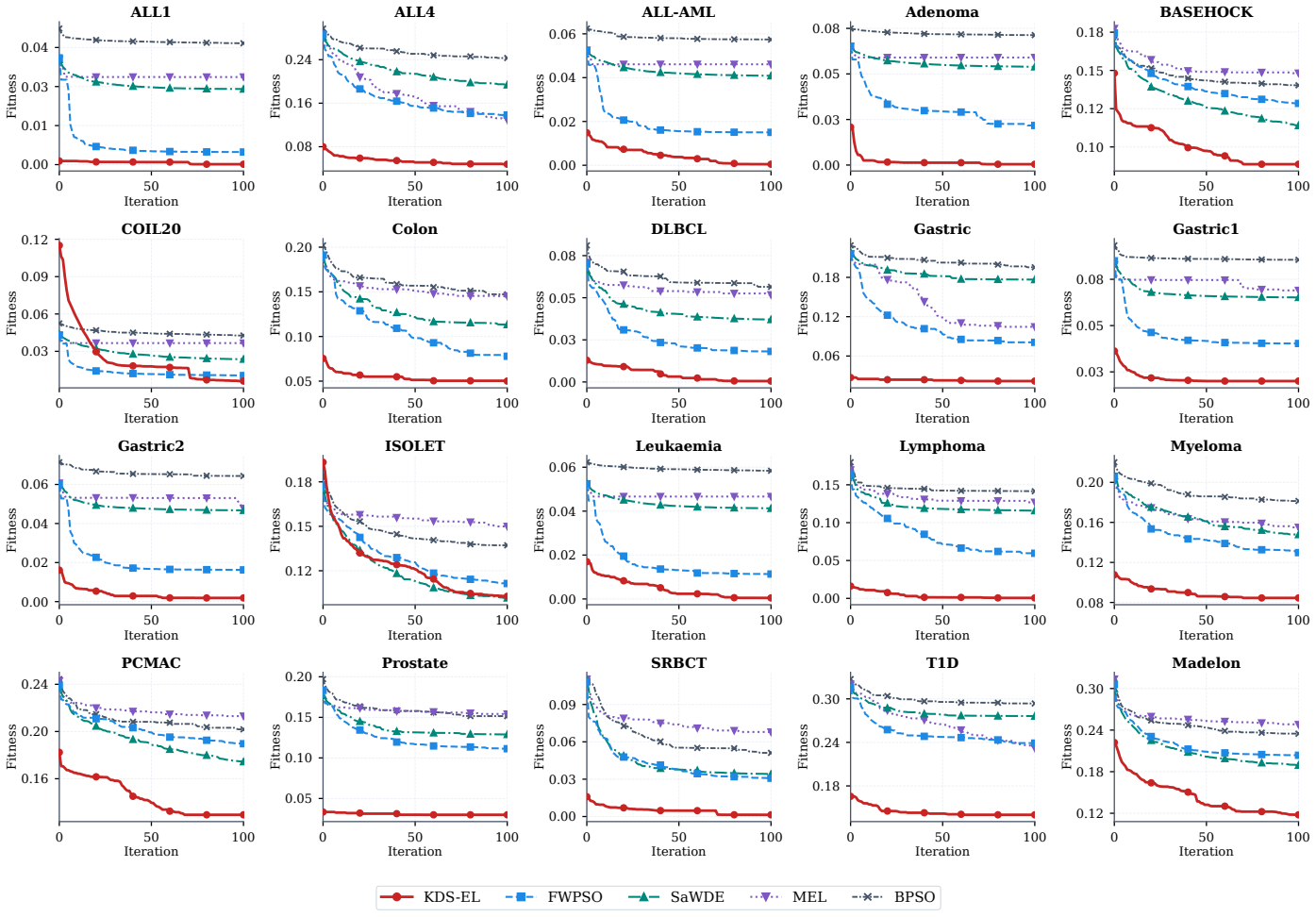


Figure 5: Convergence trajectories of mean objective fitness.

Table 6: Pairwise Wilcoxon signed-rank test results on classification accuracy across 20 benchmark datasets.

Method	KDS-EL	KDS-EL w/o TPS	KDS-EL w/o TPS&KER	MEL	bIRIME	FWPSO	PSO-CSM	BinCOA	SaWDE	IGPSO	DENCA-SO	BPSO
KDS-EL	—	12/8/0	19/1/0	20/0/0	7/5/8	19/1/0	20/0/0	18/0/2	19/1/0	20/0/0	18/1/1	20/0/0
KDS-EL w/o TPS	0/8/12	—	15/4/1	20/0/0	4/6/10	19/1/0	20/0/0	18/0/2	19/1/0	20/0/0	18/0/2	20/0/0
KDS-EL w/o TPS&KER	0/1/19	1/4/15	—	18/1/1	4/1/15	11/4/5	15/4/1	12/3/5	15/0/5	19/0/1	15/2/3	16/3/1
MEL	0/0/20	0/0/20	1/1/18	—	0/2/18	0/3/17	1/6/13	2/5/13	2/5/13	2/11/7	8/1/11	11/4/5
bIRIME	8/5/7	10/6/4	15/1/4	18/2/0	—	20/0/0	20/0/0	19/1/0	20/0/0	20/0/0	19/1/0	20/0/0
FWPSO	0/1/19	0/1/19	5/4/11	17/3/0	0/0/20	—	16/4/0	8/8/4	15/2/3	20/0/0	16/2/2	20/0/0
PSO-CSM	0/0/20	0/0/20	1/4/15	13/6/1	0/0/20	0/4/16	—	6/4/10	12/4/4	20/0/0	10/6/4	16/4/0
BinCOA	2/0/18	2/0/18	5/3/12	13/5/2	0/1/19	4/8/8	10/4/6	—	8/5/7	15/3/2	16/4/0	18/2/0
SaWDE	0/1/19	0/1/19	5/0/15	13/5/2	0/0/20	3/2/15	4/4/12	7/5/8	—	10/3/7	12/1/7	20/0/0
IGPSO	0/0/20	0/0/20	1/0/19	7/11/2	0/0/20	0/0/20	0/0/20	2/3/15	7/3/10	—	8/3/9	13/3/4
DENCA-SO	1/1/18	2/0/18	3/2/15	11/1/8	0/1/19	2/2/16	4/6/10	0/4/16	7/1/12	9/3/8	—	9/3/8
BPSO	0/0/20	0/0/20	1/3/16	5/4/11	0/0/20	0/0/20	0/4/16	0/2/18	0/0/20	4/3/13	8/3/9	—

Entries denote pairwise win/tie/loss ($+/\approx/-$) counts between the row method and column method evaluated across 20 benchmark datasets according to the paired Wilcoxon signed-rank test at $\alpha = 0.05$.

Supplementary Table S4.

A critical insight emerging from the empirical progression, visualized through multi-dataset radar profiles in Fig. 7 and convergence curves in Fig. 8, concerns asymmetric search complexity across constituent mechanisms. Transitioning from the original MEL algorithm to KDS-EL w/o TPS&KER yields the largest numerical shift in raw accuracy, producing a 4.58 pp elevation from 0.8868 to 0.9325 alongside a dimensional contraction from 2,722.2 to 290.5 features. Methodologically, this surge reflects the combined necessity of mutual information soft prior initialization and adaptive diversity recovery. In ultra-high-dimensional spaces, unguided random initialization in the original MEL algorithm indiscriminately activates approximately half of all candidate dimensions, trapping particles within the catastrophic 50% noise attractor where $d \approx D/2$. By continuous domain-relevance seeding, the MII mechanism immediately places the initial swarm within compact, discriminative subspaces, while the AIR strategy purges stagnant particles and restores momentum, eliminating over 2,400 uninformative dimensions.

Crucially, static filter-based relevance and stochastic restarts cannot navigate the combinatorial fine-tuning required in small-sample regimes: retaining 290.5 features leaves candidate subsets vulnerable to finite-sample correlation noise and the Winner's curse. The core evolutionary intelligence of KDS-EL resides in the subsequent closed-loop cascade that replaces MEL's heuristic bidirectional updates with feedback-driven refinement. Specifically, the KER strategy harvests generational fitness differentials within the importance vector $\mathbf{W}$ under exponential forgetting, steering targeted perturbations toward high-utility biomarker pairs. This closed-loop exploitation delivers an additional 1.63 pp accuracy gain while halving retained features to 121.7 in KDS-EL w/o TPS. Finally, dynamic Two-Phase Weight Scheduling temporally shifts selection pressure from coarse pruning to fine-grained discrimination, securing a further 0.86 pp gain and compressing the signature to 87.8 features. Together, the KER and TPS strategies achieve an intensive 2.49 pp accuracy breakthrough and purge over 200 redundant dimensions beyond prior seeding, confirming that KDS-EL establishes a cohesive closed-loop evolutionary sys-

Table 7: Holm-Bonferroni step-down procedure testing directional superiority of KDS-EL over all competitor methods across 20 benchmark datasets (nominal $\alpha = 0.05$).

Rank	Competitor Method	ΔAcc	W/T/L	W	p-value	α_{Holm}	Reject H_0?
1	KDS-EL w/o TPS&KER	+0.0248	19/1/0	0.0	1.91e-06	0.0045	Yes
2	MEL	+0.0706	20/0/0	0.0	1.91e-06	0.0050	Yes
3	FWPSO	+0.0396	19/1/0	0.0	1.91e-06	0.0056	Yes
4	PSO-CSM	+0.0552	20/0/0	0.0	1.91e-06	0.0063	Yes
5	IGPSO	+0.0734	20/0/0	0.0	1.91e-06	0.0071	Yes
6	BPSO	+0.0875	20/0/0	0.0	1.91e-06	0.0083	Yes
7	SaWDE	+0.0630	19/1/0	1.0	3.81e-06	0.0100	Yes
8	BinCOA	+0.0600	18/0/2	3.0	9.54e-06	0.0125	Yes
9	DENCA-SO	+0.0862	18/1/1	3.0	9.54e-06	0.0167	Yes
10	KDS-EL w/o TPS	+0.0086	12/8/0	9.0	6.29e-05	0.0250	Yes
11	bIRIME	+0.0119	7/5/8	47.0	2.96e-02	0.0500	Yes

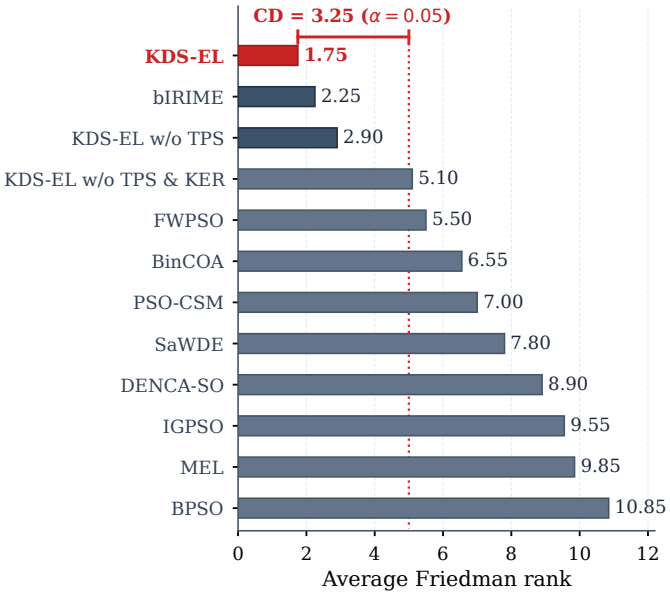


Figure 6: Average Friedman ranking distribution.

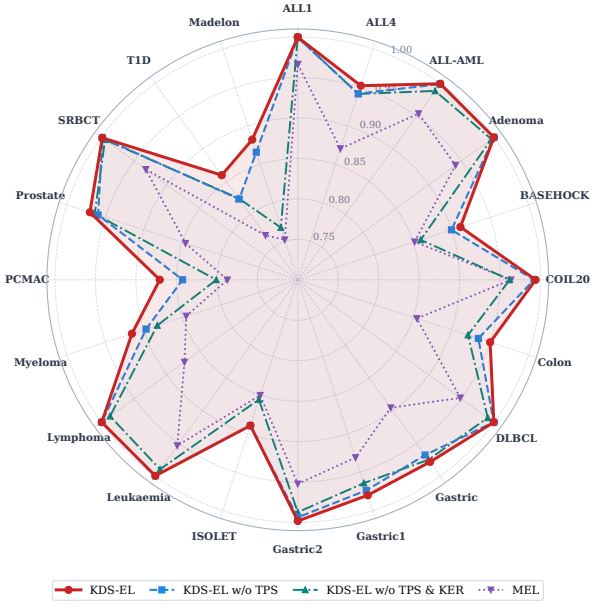


Figure 7: Multi-dataset radar visualization of classification accuracy.

Table 8: Comparison of KDS-EL and MEL performance.

Dataset	MEL		KDS-EL	
	Acc ± Std	Number	Acc ± Std	Number
ALL1	0.9664±0.0023	4,248	**0.9998±0.0001**	39
ALL4	0.8704±0.0661	1,082	**0.9522±0.0061**	124
ALL_AML	0.9539±0.0008	2,369	**0.9995±0.0002**	67
Adenoma	0.9410±0.0011	2,483	**0.9996±0.0002**	66
BASEHOCK	0.8518±0.0063	1,965	**0.9113±0.0120**	134
COIL20	0.9634±0.0017	323	**0.9939±0.0011**	84
Colon	0.8548±0.0276	702	**0.9500±0.0075**	14
DLBCL	0.9483±0.0058	2,784	**0.9995±0.0002**	64
Gastric	0.8955±0.0786	3,604	**0.9782±0.0034**	100
Gastric1	0.9310±0.0155	7,393	**0.9800±0.0003**	31
Gastric2	0.9522±0.0170	7,084	**0.9981±0.0029**	83
ISOLET	0.8501±0.0067	250	**0.8894±0.0056**	60
Leukaemia	0.9534±0.0015	2,409	**0.9995±0.0002**	66
Lymphoma	0.8731±0.0239	1,404	**0.9994±0.0003**	45
Myeloma	0.8450±0.0244	2,137	**0.9155±0.0101**	250
PCMAC	0.7870±0.0080	1,339	**0.8705±0.0139**	91
Prostate	0.8458±0.0083	5,058	**0.9701±0.0032**	26
SRBCT	0.9321±0.0085	906	**0.9986±0.0004**	64
T1D	0.7679±0.0461	6,715	**0.8598±0.0094**	330
madelon	0.7521±0.0095	190	**0.8822±0.0180**	18
Average	0.8868±0.0658	2,722.2	**0.9574±0.0482**	87.8
Δ vs. MEL	—		**+7.06 pp**	-96.8%

Entries in bold denote superior classification performance, while underlines indicate more compact feature subsets. Results are reported as Mean ± Std across 10 independent runs.

tem rather than a superficial filter pre-selector.

4.2.6. *Computational efficiency and runtime analysis*

Computational efficiency constitutes a vital evaluation dimension for evolutionary feature selection, particularly within wrapper-based metaheuristics where iterative classifier evaluations dominate wall-clock execution time. To assess real-world computational tractability, Fig. 9 visualizes the empirical trade-off distribution between classification accuracy and selected feature subset cardinality across the evaluated problem domains. Comprehensive per-dataset execution measurements across all evaluated algorithms are systematically documented in Supplementary Table S5.

As illustrated in Fig. 9 and detailed in Supplementary Table S5, KDS-EL establishes a highly favorable efficiency profile, achieving an omnibus average execution time of 51.3 seconds across the benchmark suite and delivering a 3.4-fold acceleration over the original MEL algorithm. While evolutionary filter paradigms such as DENCA-SO [14] exhibit rapid execution by evaluating candidate subsets through algebraic surrogate metrics rather than inductive classifiers, they do so at the cost of substantial predictive deficits and severe feature bloat. Conversely, contemporary state-of-the-art wrapper optimizers, including bIRIME [13] and BinCOA [15], suffer from steep computational penalties that escalate dramatically on large-scale and high-dimensional problem instances. Standard swarm architectures such as PSO-CSM [11], SaWDE [9], and BPSO [7] likewise incur prolonged execution times because unguided stochastic updates wander extensively across dense combinatorial spaces. In contrast, KDS-EL consistently occupies the ideal high-accuracy, ultra-sparse, and low-latency region in the upper-left quadrant of Fig. 9, achieving practical scalability without sacrificing predictive precision.

The pronounced computational acceleration of KDS-EL stems directly from the closed-loop synergy of its algorithmic mechanisms, which fundamentally reduce the evaluation cost formalized in Eq. (24). The primary computa-

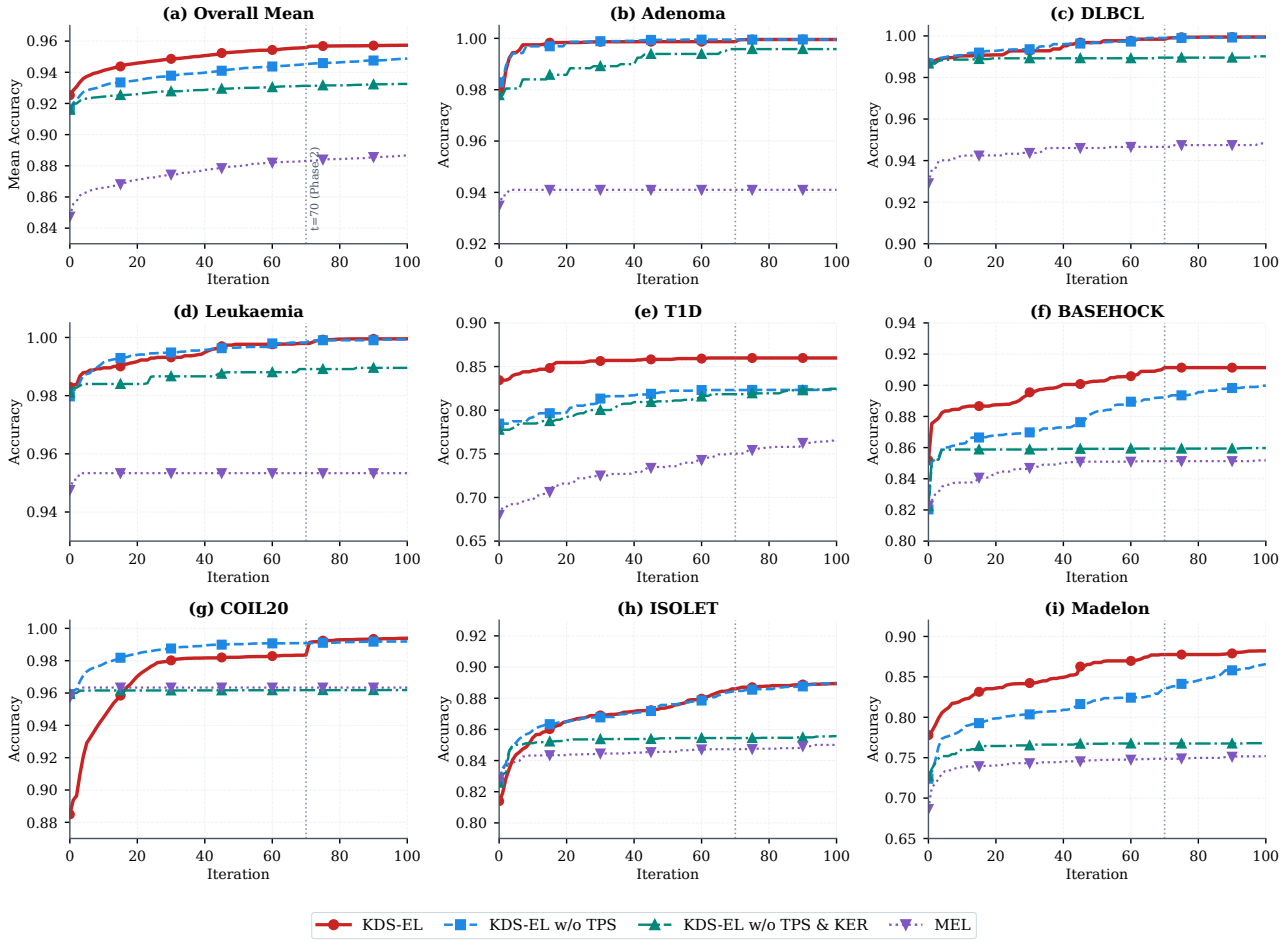


Figure 8: Convergence trajectories of classification accuracy before and after algorithm improvement.

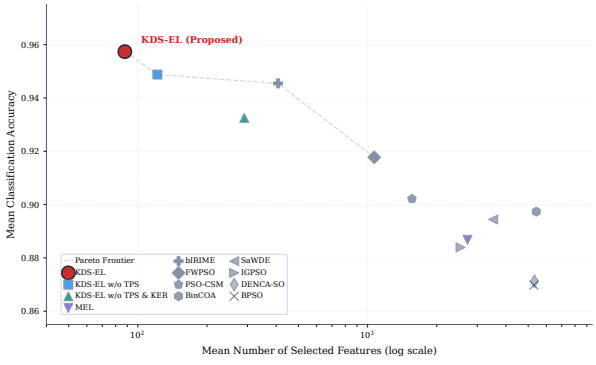


Figure 9: Trade-off distribution between classification accuracy and selected feature cardinality.

tional bottleneck in wrapper feature selection resides in repeated distance evaluations, where the cost of computing pairwise Euclidean distances under k-NN scales linearly with the active feature cardinality d_{selected}. By combining prior-guided initialization with aggressive early-stage feature penalization in Two-Phase Weight Scheduling, KDS-EL contracts candidate subsets from thousands of raw dimensions down to compact representations within early generations, thereby diminishing the per-evaluation classifier latency by orders of magnitude. Furthermore, rather than generating multiple trial vectors per agent as in complex multi-operator metaheuristics, the KER strategy confines targeted local exploitation to a bounded elite subset under $k_e \ll N$, where candidate mutations are admitted solely through the greedy in-place replacement rule in Eq. (21). This architectural restriction prevents redundant classifier invocations and maintains generational complexity strictly linear in population size, harmonizing rigorous exploration depth with optimal computational thrift.

4.2.7. *Parameter sensitivity and robustness analysis*

To evaluate the algorithmic robustness of KDS-EL and justify its default parameter configurations, a one-factor-at-a-time parameter sensitivity analysis is conducted following the standard evaluation methodology in evolutionary machine learning [34]. The empirical assessment is carried out across four representative benchmark datasets spanning distinct dimensional scales and sample sizes: *Adenoma*, *Colon*, *ALL4*, and *BASEHOCK*. Five key hyperparameters governing the core algorithmic strategies are systematically investigated: the knowledge weight decay factor $\gamma \in (0, 1)$, the phase split ratio $\kappa \in (0, 1)$, the elite perturbation fraction $\phi \in (0, 1)$, the mutual information initialization ratio $\rho \in (0, 1)$, and the elite pool size $k_e \in \mathbb{N}^+$, while all remaining parameters are held at their default values defined in Table 3. Table 9 summarizes the tested parameter ranges, recommended optimal configurations, stable operating bands, and maximum accuracy deviations across the four representative benchmarks. Furthermore, Fig. 10 illustrates the corresponding classification accuracy and feature-count trajectories alongside stable operating bands across all five parameters, with complete per-dataset evaluation matrices provided in Supplementary Table S6.

Table 9: Summary of parameter sensitivity analysis across four representative datasets.

Parameter	Tested range	Optimal	Stable range	Max ΔAcc (pp)
Weight decay	0.90–1.00	0.99	[0.95, 0.995]	±1.2
Phase split	0.3–0.9	0.70	[0.60, 0.80]	±0.7
Flip fraction	0.005–0.10	0.01	[0.005, 0.02]	±1.3
MI init ratio	0.0–1.0	0.50	[0.25, 0.75]	±1.2
Elite count	1–10	3	[2, 5]	±0.9

Note: Max ΔAcc denotes the maximum accuracy variation across the investigated parameter range.

The empirical sensitivity trajectories reveal that KDS-EL maintains stable optimization dynamics through coordinated phase scheduling and prior guid-

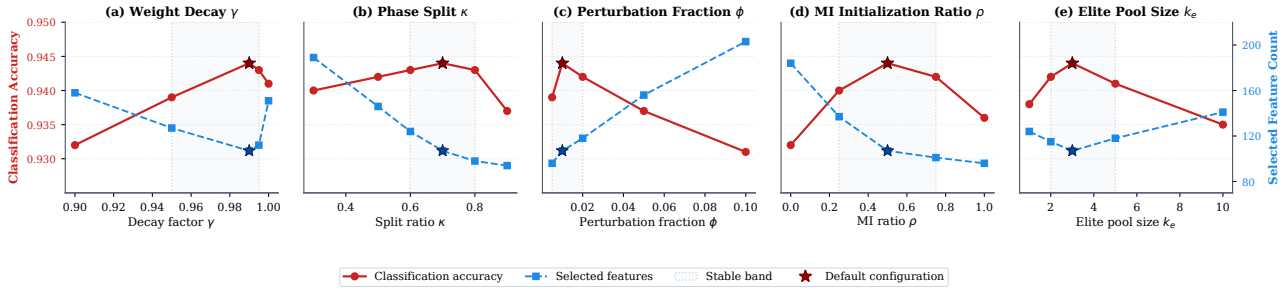


Figure 10: Parameter sensitivity and stability profiles of KDS-EL across five key hyperparameters.

ance. The phase split ratio κ governs the temporal transition between dimensional pruning in Phase 1 and classification refinement in Phase 2 via Eq. (22). Allocating an insufficient budget to Phase 1 with $\kappa < 0.50$ leaves the swarm encumbered by redundant dimensions; conversely, an overly prolonged Phase 1 with $\kappa > 0.85$ curtails the exploitation horizon necessary for uncovering subtle synergies. The stable interval $\kappa \in [0.60, 0.80]$ confirms that default $\kappa = 0.70$ balances dimensional contraction and precision. Concurrently, the mutual information initialization ratio ρ regulates the trade-off between prior knowledge and exploratory coverage in Eq. (12). Omitting priors when $\rho \to 0$ forces unguided exploration, whereas excessive seeding with $\rho \to 1.0$ homogenizes candidate masks and provokes premature entrapment. The operating band $\rho \in [0.25, 0.75]$ with default $\rho = 0.50$ robustly sustains population diversity while seeding decisive early momentum.

Regarding memory accumulation and local exploitation, sensitivity trajectories confirm that closed-loop learning is intrinsically resilient across wide parameter ranges. The weight decay factor γ in Eq. (17) governs historical retention in $\mathbf{W}$. Aggressive decay with $\gamma < 0.95$ induces premature search amnesia, while unattenuated accumulation at $\gamma = 1.00$ causes obsolete feedback to persist indefinitely. The default setting $\gamma = 0.99$, situated within the stable band $[0.95, 0.995]$, effectively harmonizes knowledge retention with adaptive sensitivity to emergent synergies. Furthermore, perturbation fraction ϕ and pool size k_e calibrate local search within the KER strategy. Setting ϕ excessively large introduces disruptive alterations that fail the greedy pairwise in-place acceptance criterion in Eq. (21); conversely, the default setting $\phi = 0.01$ within the stable band $[0.005, 0.02]$ provides controlled perturbation that maximizes improving substitutions. Bounding elite pool size to $k_e = 3$ within the stable interval $[2, 5]$ concentrates computation on high-performing exemplars while strictly limiting generational overhead to $k_e \ll N$.

Synthesizing across all five hyperparameters, performance trajectories in Fig. 10 consistently exhibit wide, forgiving basins of attraction. Across identified stable operating bands, maximum accuracy variation remains strictly within 1.5%, and KDS-EL maintains robust directional superiority over competitor algorithms throughout the tested spectrum, confirming plug-and-play adaptability across heterogeneous problem domains.

4.2.8. Discussion

A critical empirical phenomenon in high-dimensional biomedical feature selection is the accuracy ceiling effect: across eight microarrays, classification accuracy exceeds 0.998 across leading optimizers. Under such near-perfect separation, accuracy differences narrow below 0.002, reflecting variations in at most a single sample. Consequently, classification accuracy ceases to provide sufficient discriminative resolution, making active feature cardinality decisive for assessing model parsimony [2, 3].

Under this saturated regime, KDS-EL exhibits marked superiority in feature compactness, isolating ultra-sparse biomarker signatures of only 39 to 83 features across all eight ceiling benchmarks. In stark contrast, competing metaheuristics, evolutionary filters, and the original MEL algorithm retain hundreds to thousands of redundant genes. This divergence highlights a pervasive failure mode: lacking early-stage dimensional penalization and elite refinement, optimizers succumb to feature bloat once high accuracy is attained. From a translational perspective, contracting candidate signatures to 40 to 80 biomarkers scales down the hypothesis space into a tractable candidate regime for prospective clinical validation, whereas uncurated signatures spanning thousands of dimensions remain prohibitive for downstream wet-lab screening.

Beyond nearest-neighbor metrics, a fundamental concern is inductive bias coupling [4]: whether optimized subsets merely overfit the 3-NN surrogate. To examine classifier-agnostic generalizability, compact subsets discovered by KDS-EL were transferred to Support Vector Machines and Random Forest classifiers. As documented in Supplementary Table S9, compact subsets deliver substantial classification gains across all architectures: on the 12,625-dimensional *ALL4* benchmark, contracting features by over 99.0% elevates all classifier accuracies, while on *Adenoma*, all three classifiers reach 100% accuracy on 66 selected dimensions, confirming that KDS-EL extracts intrinsically salient, generalizable features.

Furthermore, translating computational feature selection into clinical insight requires that isolated dimensions reflect genuine pathophysiological mechanisms rather than statistical noise. Granular inspection reveals strong concordance with established oncological biology, detailed in Supplementary Table S10. In *Colon* adenocarcinoma, top selected features map directly to drivers of carcinogenesis: KDS-EL isolates *COL11A1*, a recognized biomarker of desmoplastic stroma and tumor invasion [42]; *GUCA2A*, a mucosal differentiation hormone marking early adenoma progression [43]; and smooth muscle structural markers *DES* and *ACTG2*. Similarly, in *Prostate* carcinoma, KDS-EL isolates *HPN* and *AMACR*, established immunohistochemical gold standards for diagnosis [44, 45], corroborating that KDS-EL captures authentic drivers of malignancy.

5. Conclusion

In this paper, KDS-EL is proposed to overcome search blindness, evolutionary amnesia, destructive perturbation, and static weighting bottlenecks in ultra-high-dimensional wrapper feature selection. By integrating mutual information prior initialization with adaptive recovery, KDS-EL establishes informative search basins while safeguarding diversity. To eliminate destructive mutations, the KER strategy harvests fitness differentials under exponential memory decay, steering targeted refinement within bounded computation. Furthermore, dynamic Two-Phase Weight Scheduling temporally decouples coarse dimensional pruning from fine classification refinement. Benchmark evaluations across 20 datasets demonstrate that KDS-EL establishes leading performance under multi-criteria evaluation, attaining 0.9574 mean accuracy, contracting feature subsets to an average of 87.8 dimensions, and delivering a 3.4-fold acceleration over the original MEL algorithm.

Beyond nearest-neighbor metrics, extensive evaluations confirm that the discovered compact biomarker signatures exhibit robust classifier-agnostic generalizability across Support Vector Machines and Random Forest models. Under saturated regimes, KDS-EL suppresses feature bloat, isolating compact signatures of 39 to 83 biomarkers mapping directly to verified oncological drivers, including *COL11A1*, *GUCA2A*, *HPN*, and *AMACR*. Methodologically, the synergy of knowledge-guided importance accumulation, bounded elite refinement, and dynamic objective scheduling provides a versatile foundational paradigm for continuous-discrete swarm intelligence.

Several promising trajectories emerge for future exploration. First, integrating self-supervised embeddings could discover higher-order epistatic interactions across massive cohorts. Second, cross-cohort transfer mechanisms could enable reusing accumulated importance memories. Third, evaluating cross-run feature selection stability alongside prospective multi-center cohort validation will be essential to advance computational biomarker signatures toward clinical translation.

CRediT authorship contribution statement

Anonymous Author: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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